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INTERNATIONAL MARITIME ORGANIZATION

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SUB-COMMITTEE ON CARRIAGE OF CARGOES AND CONTAINERS
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Agenda item 18

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DRAFT REPORT TO THE MARITIME SAFETY COMMITTEE AND THE MARINE ENVIRONMENT PROTECTION COMMITTEE

1 GENERAL

Introduction

1.1 The Sub-Committee on Carriage of Cargoes and Containers (CCC), chaired by Ms. M. Adams (Marshall Islands), held its eighth session from 14 to 23 September 2022. The Vice-Chair, Mr. D. Anderson (Australia), was also present.

1.2 The session was attended by delegations from Member States and Associate Members of IMO; and by observers from intergovernmental organizations and non-governmental organizations in consultative status, as listed in document CCC 8/INF.1.

Secretary-General's opening address

1.3 The Secretary-General welcomed participants and delivered the opening address. In his opening speech, the Secretary-General expressed condolences to King Charles III and other members of the Royal Family, the Government, the people of the United Kingdom of Great Britain and Northern Ireland, and the Commonwealth on the death of Her Majesty, Queen Elizabeth II. A moment of silence in memory of the Queen was observed by the Sub-Committee and many delegations also made statements to convey their condolences.

1.4 The Secretary-General also expressed his condolences to the family and colleagues of the former Chair of the Maritime Safety Committee, Mr. Christian Breinholt (Denmark), who recently passed away (see also paragraph 17.[…]). The full text of the Secretary-General's

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opening address can be downloaded from the IMO website at the following address: https://www.imo.org/en/MediaCentre/SecretaryGeneral/Pages/Secretary-GeneralsSpeechesToMeetings.aspx

Chair's remarks

1.5 In response, the Chair thanked the Secretary-General for his words of guidance and encouragement and assured him that his advice and requests would be given every consideration in the deliberations of the Sub-Committee. On behalf of the Sub-Committee, the Chair also expressed condolences to the members of the Royal Family, the Government, the people of the United Kingdom of Great Britain and Northern Ireland, and the Commonwealth on the death of Her Majesty, Queen Elizabeth II.

Adoption of the agenda and related matters

1.6 The Sub-Committee adopted the agenda (CCC 8/1/Rev.1) and agreed to be guided in its work, in general, by the annotations contained in document CCC 8/1/1 (Secretariat) and the working arrangements in document CCC 8/1/2 (Chair).

Use of hybrid meeting capabilities

1.7 The Sub-Committee recalled that the 127th session of the Council, held remotely from 11 to 25 July 2022, agreed to (C 127/D, paragraph 17.3):

.1 use hybrid meeting capability to complement in-person meetings from September 2022, for a trial period of one year;

.2 apply the Rules of Procedure of the Council and the Interim Guidance to facilitate remote sessions of the IMO Council during the COVID-19 pandemic, as appropriate, and agreed that only representatives of the Members attending the meeting in person at IMO Headquarters would be allowed to vote; and

.3 invite other organs of the Organization to follow the above decisions of the Council and to report to a future session of the Council on their experience with such meetings.

1.8 In this context, noting the Committees will be held after CCC 8 and after the consultation with the MSC Chair, the Sub-Committee agreed to use hybrid meeting capabilities to complement in-person meetings (hereinafter "hybrid meetings") for this session as a trial; and report to the Committees on the experience of hybrid meetings, if any.

2 DECISIONS OF OTHER IMO BODIES

General

2.1 The Sub-Committee noted the outcomes of MSC 104, MSC 105, MEPC 77, MEPC 78, C/ES.34 and A 32, relevant to the work of the Sub-Committee, as reported in documents CCC 8/2 and CCC 8/2/1 (Secretariat), and took them into account in its deliberations when dealing with the relevant agenda items.

2.2 The Sub-Committee also noted that MSC 104 had agreed to an ad hoc midterm amendment cycle under exceptional circumstances in accordance with the *Guidance on entry into force of amendments to the 1974 SOLAS Convention and related mandatory instruments* (MSC.1/Circ.1481), i.e. entry into force on 1 January 2026 for the draft amendments to SOLAS and related mandatory instruments under the purview of the Committee adopted before 1 July 2024 (MSC 104/18, paragraph 3.16.1).

2.3 Having further noted that MSC 105 requested the CCC Sub-Committee to consider documents A 32/12/2 and MSC 105/2/2, and matters related to the development of safety requirements to support the achievement of the decarbonization goal, in the context of its work on the development of the IGF Code, the Sub-Committee decided to consider document CCC 8/2/1 (IACS) under agenda item 3 (see paragraphs 3.[…] to 3.[…]).

2.4 In addition to the above information, the outcome of:

.1 the eighth session of the Sub-Committee on Implementation of IMO Instruments (III), held from 25 to 29 July 2022, in relation to the lessons learned and preventive measures identified from accidents involving entering enclosed spaces aboard, was considered under agenda item 8 (see paragraphs 8.[…] to 8.[...]); and

.2 C 127 on the use of hybrid meeting capabilities was considered under agenda item 1 (see paragraphs 1.[…] to 1.[...]).

Ongoing military conflict between the Russian Federation and Ukraine; and the Black Sea Grain Initiative

2.5 The Sub-Committee noted a statement made by the delegation of Ukraine, as set out in annex […], pointing out the impact of the ongoing Russian Federation's invasion of Ukraine on the global food crisis and reporting the results of the implementation of the Initiative on

the Safe Transportation of Grain and Foodstuffs from Ukrainian Ports (Black Sea Grain Initiative), which was launched on 22 July 2022. In particular, the delegation confirmed Ukraine's determination to ensure effective implementation of the Black Sea Grain Initiative; expressed its appreciation to the IMO Secretariat for its significant contribution to the preparation of this Agreement; and emphasized that the successful implementation of the Black Sea Grain Initiative is overshadowed by the ongoing theft of Ukrainian agricultural products by the Russian occupying forces on the territory of Ukraine, as well as the attacks on port infrastructure used within the framework of the Black Sea Grain Initiative.

2.6 In this regard, the Secretary-General provided an update on IMO's active role in the Black Sea Grain Initiative, in particular expressing gratitude to all parties that are contributing to the successful operation of the Initiative, which he observed first-hand during his recent visit to the port of Odesa.

2.7 The delegation of the Russian Federation made a statement, as set out in annex […], commenting on the statement made by the delegation of Ukraine and on other matters raised during the discussion.

2.8 The delegations of Australia, The Bahamas, France (supported by Belgium, Croatia, Denmark, Estonia, Finland, Germany, Greece, Ireland, Italy, Netherlands, Norway, Poland, Portugal, Spain, Sweden and the European Commission), Canada, Japan, Türkiye, United Kingdom and United States also made statements as set out in annex […], expressing their solidarity with Ukraine. The majority of delegations emphasized that Russia and its unprovoked war against Ukraine was in contravention of the Charter of the United Nations and the purposes of IMO, as set forth in Article 1 of the Convention. They observed that Russia alone bore full responsibility for the increasingly grave consequences of its war, including the need for a negotiated Black Sea Grain Initiative to allow Ukraine to export its grain from its ports. Many delegations once again called on the Russian Federation to immediately withdraw its forces from Ukraine, including its territorial waters.

3 AMENDMENTS TO THE IGF CODE AND DEVELOPMENT OF GUIDELINES FOR LOW-FLASHPOINT FUELS

BACKGROUND

3.1 The Sub-Committee recalled that CCC 7 had re-established the Correspondence Group on Development of Technical Provisions for the Safety of Ships using Low-flashpoint Fuels, to continue the work on the draft interim guidelines on safety of ships using LPG fuels and the consideration on low-flashpoint oil fuels.

REPORT OF THE CORRESPONDENCE GROUP

3.2 The Sub-Committee had for its consideration the report of the Correspondence Group on Development of the Interim Guidelines LPG, Hydrogen, Low-flashpoint Oil Fuels and Amendments to the IGF Code (CCC 8/3). The Sub-Committee noted that the Group had been organized in two parts and the second part on collecting information on the use of ammonia as fuel was submitted under agenda item 13 (CCC 8/13).

Safety provisions for ships using low-flashpoint oil fuels

3.3 The Sub-Committee noted the Group's discussion on the development of safety provisions for ships using low-flashpoint oil fuels. In this context, the Sub-Committee endorsed the Group's suggestion to develop interim guidelines for the use of oil fuels with a flashpoint between 52°C and 60°C, covering oil-based fossil fuels, synthetic fuels, biofuels and any mixture thereof, and taking into account the deliberations of the Group in paragraphs 4 to 8 and documents CCC 4/INF.11, CCC 6/3/2, CCC 6/3/7, CCC 6/INF.6, CCC 6/WP.3, CCC 7/3, and annex 1 to document CCC 8/3.

Interim guidelines for ships using LPG fuels

3.4 The Sub-Committee noted the discussions and progress made by the Group regarding the development of the interim guidelines for ships using LPG as fuel. Having noted the support expressed on the development of the interim guidelines for ships using LPG, the Sub-Committee instructed the Working Group on Development of Technical Provisions for Safety of Ships Using Alternative Fuels, if established, to finalize the draft interim guidelines for ships using LPG fuels, based on annexes 2 and 3 to document CCC 8/3, as a matter of priority.

Interim guidelines for ships using hydrogen as fuel

3.5 The Sub-Committee noted the discussions and progress made by the Group on the development of guidelines for the safety of ships using hydrogen as fuel. Following discussion, the Sub-Committee also noted the following views expressed on the matter:

.1 the application experience of hydrogen engines is quite insufficient since there is no marine hydrogen internal combustion engine at present and a cautious approach should be taken;

.2 the development of provisions related to hydrogen should be technology neutral;

.3 emergency shutdown (ESD) concept should not be excluded at this stage; and

.4 the suitability of the gas safe machinery spaces should also be considered.

3.6 In this context, the Sub-Committee instructed the Working Group on Development of Technical Provisions for Safety of Ships Using Alternative Fuels, if established, to further develop the draft interim guidelines for ships using hydrogen as fuel, based on annex 9 to document CCC 8/3.

Development of safety requirements to support achievement of a decarbonization goal and work plan for the future development of low-flashpoint fuels

3.7 The Sub-Committee recalled that MSC 105 had instructed CCC 8 to further consider documents A 32/12/2 and MSC 105/2/2, and matters related to the development of safety requirements to support the achievement of the decarbonization goal, in the context of its work on the development of amendments to the IGF Code and safety provisions for alternative fuels, and advise the Committee on how best to proceed (MSC 105/20, paragraph 14.11.3).

3.8 The Sub-Committee also recalled that MSC 102 had changed the target completion year of the existing output on "Amendments to the IGF Code and development of guidelines for low-flashpoint fuels" to "continuous", taking into account the work plan related to the development of safety provisions for several alternative fuels under the IGF Code.

3.9 In this regard, the Sub-Committee further recalled that both MSC 104 and MEPC 76 had noted the work plan prepared by the CCC Sub-Committee for the development of new low-flashpoint fuels under the IGF Code, which included ongoing and planned development of safety provisions on several alternative fuels.

3.10 The Sub-Committee had for its consideration the following documents:

.1 A 32/12/2 (IACS), considering the role of MSC in addressing the multidimensional challenge posed by the decarbonization ambition and the development of the necessary detailed requirements and regulations to deliver a safe zero-CO_2-emitting ship;

.2 MSC 105/2/2 (IACS), considering aspects of an approach and offers preliminary views on the risks associated with the options currently researched and trialled to deliver a safe zero-CO_2-emitting ship; procedural and administrative aspects due to the governance of the regulatory development process at IMO, where the agreement on a new output needs to be sought and agenda items would be allocated to future sessions of the Sub-Committees; suggesting that MSC consider having a standing agenda item on its agenda, under which the Committee will receive, evaluate and prioritize potential solutions; and

.3 CCC 8/2/1 (IACS), following up on documents to A 32 and MSC 105, offering further comments to articulate the need for a broad approach to regulating safety of ships with new technologies and alternative fuels to support the achievement of IMO's decarbonization goals. It considers that the advice to MSC needs to recognize the broad spectrum and complexity of issues without necessarily confining that advice to the context of the Sub-Committee's work on the development of the IGF Code and safety provisions for alternative fuels.

3.11 In the ensuing discussion, the Sub-Committee noted the following views expressed on the matter:

.1 the development of safety requirements for alternate fuels and technologies is a broad and complex matter with many aspects involved;

.2 MSC should have a leading and active role on the development of safety requirements to support the achievement of a decarbonization goal;

.3 a holistic approach is needed for the development of safety requirements to support the achievement of a decarbonization goal and not all issues can be solved by the CCC Sub-Committee itself;

.4 the development of safety requirements to support the achievement of a decarbonization goal should be technology neutral, to ensure all fuels be treated equally;

.5 the work should not be confined in the scope of the IGF Code and matters such as training of seafarers, toxicity of the fuels and environmental aspects should also be dealt with;

.6 the additional resources, e.g. eight days of CCC meeting and parallel working groups, to accelerate the development of safety requirements for alternate fuels are welcomed and the Organization is on the right course to deliver legal certainties to the industry;

.7 at this stage, there is no need to alter the existing arrangements for development of safety requirements for alternate fuels and the CCC Sub-Committee is the right body for handling this matter, with demonstrated smooth cooperation and efficiency;

.8 implementation and communication should be enhanced and IMO's technical cooperation programme could be unitized; and

.9 the establishment of a specific agenda item for the development of safety requirements for alternate fuels and technologies at MSC is necessary.

3.12 Having noted the overwhelming support for documents A 32/12/2, MSC 105/2/2 and CCC 8/2/1, the Sub-Committee:

.1 thanked IACS for its submissions, in particular for raising the important aspect on development of safety requirements to support the achievement of a decarbonization goal;

.2 recognized the wide variety and complexity of the matter on the development of safety requirements for alternate fuels and the technologies it may involve, in particular that the work may not necessarily be confined to the context of the IGF Code;

.3 noted that the CCC Sub-Committee, as a subsidiary body under both MSC and MEPC, has been working on a list of alternative fuels in accordance with its work plan;

.4 noted that there may be a need to adjust the existing output of "Amendments to the IGF Code and development of guidelines for low-flashpoint fuels", with a view to streamlining the procedure and clarifying its scope;

.5 reiterated the Sub-Committee's commitment to continue ensuring appropriate coordination with other sub-committees, noting the variety and complexity of issues that will need to be addressed in the regulatory development process;

.6 noted that there may be a need to recommend to MSC to consider establishing an output, with a view to coordinating its work with MEPC and other committees, noting that there is an existing output on development of further measures to enhance the safety of ships relating to the use of fuel oil; and

.7 instructed the Working Group on Development of Technical Provisions for Safety of Ships using Alternative Fuels to:

.1 further consider documents CCC 8/2/1, A 32/12/2 and MSC 105/2/2, and matters related to the development of safety requirements to support the achievement of a decarbonization goal and advise the Sub-Committee on how best to proceed; and

.2 update the work plan for the development of the IGF Code and safety provisions for alternative fuels, based on annex 2 to document CCC 7/15.

ESTABLISH THE WORKING GROUP ON DEVELOPMENT OF TECHNICAL PROVISIONS FOR SAFETY OF SHIPS USING ALTERNATIVE FUELS

3.13 The Sub-Committee agreed to establish the Working Group on Development of Technical Provisions for Safety of Ships Using Alternative Fuels and instruct it, taking into account the comments made and decisions taken in plenary, to:

.1 finalize the draft interim guidelines for ships using LPG fuels, based on annexes 2 and 3 to document CCC 8/3;

.2 further develop the draft interim guidelines for ships using hydrogen as fuel, based on annex 9 to document CCC 8/3;

.3 develop interim guidelines for the use of oil fuels with a flashpoint between 52°C and 60°C, covering oil-based fossil fuels, synthetic fuels, biofuels and any mixture thereof, and taking into account documents CCC 4/INF.11, CCC 6/3/2, CCC 6/3/7, CCC 6/INF.6, CCC 6/WP.3, CCC 7/3, and annex 1 to document CCC 8/3;

.4 further consider documents CCC 8/2/1, A 32/12/2 and MSC 105/2/2, and matters related to the development of safety requirements to support the achievement of a decarbonization goal and advise the Sub-Committee on how best to proceed;

.5 update the work plan for the development of the IGF Code and safety provisions for alternative fuels, based on annex 2 to document CCC 7/15; and

.6 consider whether it is necessary for the correspondence group to be re-established and, if so, prepare terms of reference for consideration by the Sub-Committee.

REPORT OF THE WORKING GROUP ON DEVELOPMENT OF TECHNICAL PROVISIONS FOR SAFETY OF SHIPS USING ALTERNATIVE FUELS

3.14 Having considered the report of the Working Group on Development of Technical Provisions for Safety of Ships Using Alternative Fuels (CCC 8/WP.3), the Sub-Committee approved it in general and took action as described in paragraphs ….

[to be prepared by the Secretariat in consultation with the Chair after the session, based on the group's report and the actions requested therein, taking into account the decisions taken by the Sub-Committee during subsequent discussions]

Amendments to the IGF Code

A set of draft amendments to the IGF Code

3.15 The Sub-Committee noted the discussion on draft amendments to paragraphs 9.6, 9.6.1, 11.6.2, 9.4.7, 12.5, 13.3 and 6.7.3.1.1 of the IGF Code (paragraphs 13 to 15 and annex 4 to document CCC 8/3). In the ensuing discussion, the Sub-Committee noted the following views expressed on the matter:

.1 with regard to the draft amendments to paragraph 12.5 of the IGF Code, that definitions for small or large volumes of gas should be developed first, before the finalization of the draft amendments to the area classification in 12.5; and

.2 draft amendments to section 13 on ventilation required for safe operation of gas-fuelled machinery and equipment should be further considered by the Working Group.

3.16 Following discussion, the Sub-Committee instructed the Working Group on Amendments to the IGF Code and Review of the IGC Code to finalize the draft set of amendments to paragraphs 9.6, 9.6.1, 11.6.2, 9.4.7, 12.5, 13.3 and 6.7.3.1.1 of the IGF Code, based on annex 4 to document CCC 8/3.

3.17 Furthermore, having recalled the draft amendments prepared by the Working Group during CCC 6, i.e. annex 3 to CCC 6/WP.3, the Sub-Committee instructed the Working Group on Amendments to the IGF Code and Review of the IGC Code to finalize the draft amendments to paragraphs 9.4.8, 11.3.1 and 15.4.1.3 of the IGF Code, based on annex 3 to document CCC 6/WP.3.

Draft unified interpretation regarding paragraph 9.2.2 of the IGF Code

3.18 Having noted the discussion of the Group, the Sub-Committee requested the Secretariat to prepare the relevant draft MSC circular on the unified interpretation regarding paragraph 9.2.2 of the IGF Code, based on annex 5 to document CCC 8/3, for consideration by the Sub-Committee (see also paragraph 12.[…]).

Draft amendments to paragraph 9.3.1 of the IGF Code

3.19 Having noted the discussion on draft amendments to paragraph 9.3.1 of the IGF Code, the Sub-Committee instructed the Working Group on Amendments to the IGF Code and Review of the IGC Code, if established, to finalize the draft amendments to paragraph 9.3.1 of the IGF Code, based on annex 6 to document CCC 8/3.

Draft amendments to paragraphs 4.2.2 and 8.4.1 to 8.4.3 of the IGF Code

3.20 Having noted the discussion on draft amendments to paragraphs 4.2.2 and 8.4.1 to 8.4.3 of the IGF Code, the Sub-Committee instructed the Working Group on Amendments to the IGF Code and Review of the IGC Code, if established, to finalize the draft amendments to paragraphs 4.2.2 and 8.4.1 to 8.4.3 of the IGF Code, based on annex 7 to document CCC 8/3.

Draft amendments to part A-1, paragraphs 5.12.1, 6.9.1.1, 9.8.1, 9.8.2 and 9.8.4 and to part C-1, paragraph 18.4.1.1.1 of the IGF Code

3.21 Having noted the discussion on draft amendments to part A-1, paragraphs 5.12.1, 6.9.1.1, 9.8.1, 9.8.2, 9.8.4 and to part C-1, paragraph 18.4.1.1.1 of the IGF Code, the Sub-Committee instructed the Working Group on Amendments to the IGF Code and Review of the IGC Code, if established, to finalize the draft amendments to part A-1, paragraphs 5.12.1, 6.9.1.1, 9.8.1, 9.8.2 and 9.8.4 and to part C-1, paragraph 18.4.1.1.1 of the IGF Code, based on annex 8 to document CCC 8/3.

New submissions in relation to the amendments to the IGF and IGC Codes

3.22 The Sub-Committee had for its consideration documents:

.1 CCC 8/3/1 (IACS), proposing amendments to paragraph 5.11.2.2 of the IGC Code and paragraph 7.3.2.1 of part A-1 of the IGF Code, in particular that the value to be taken for "a" can be misinterpreted, considering that the formula and the description of "a" under the formula does not specify that it is the absolute value of manufacturing tolerance that is to be used; a similar text in paragraph 5.11.2.2 of the IGC Code should also be amended;

.2 CCC 8/3/2 (IACS), proposing a revision of regulation 11.3.3.1.2 of part A-1 of the IGF Code, as amended by resolution MSC.458(101), in particular that a difference will arise between the two types of fuel tanks in that the minimum distance for the vacuum insulation double wall tank does not include the insulation space, but the minimum distance for single wall fuel tanks includes the insulation system;

.3 CCC 8/3/3 (IACS), providing a clarification regarding regulations 11.3.2 and 11.3.3 of the IGF Code, as to the application of "A-60" fire integrity rating to the accommodation block and the deckhead of a machinery space, other than category "A" machinery space, where the deckhead is beneath the fuel containment system; and

.4 CCC 8/3/4 (Republic of Korea), providing proposals for corrections, mainly editorial to the IGF Code, along with their corresponding rationales.

3.23 In the ensuing discussion, the Sub-Committee noted the comments expressed with regard to the proposed amendments in document CCC 8/3/4, in particular that the application scope should be carefully considered by the Working Group. Subsequently, the Sub-Committee instructed the Working Group on Amendments to the IGF Code and Review of the IGC Code to further consider documents CCC 8/3/1, CCC 8/3/2, CCC 8/3/3 and CCC 8/3/4, with a view to preparing amendments to the IGF and IGC Codes.

Clarifications of fire protection requirements in the IGF Code

3.24 The Sub-Committee considered document CCC 8/17/2 (IACS), seeking clarifications of the application of paragraphs 11.3.1 and 11.3.2 of the IGF Code regarding fire protection of fuel preparation rooms facing fuel tank(s) on deck. Having noted clarifications were needed for the application of "A-60" insulation, as required by paragraph 11.3.2 of the IGF Code to the boundaries of the fuel preparation rooms, the Sub-Committee instructed the Working Group on Amendments to the IGF Code and Review of the IGC Code, if established, to further consider document CCC 8/17/2, and advise the Sub-Committee how best to proceed.

ESTABLISHMENT OF A WORKING GROUP ON AMENDMENTS TO THE IGF CODE AND REVIEW OF THE IGC CODE

3.25 The Sub-Committee agreed to establish the Working Group on the Amendments to the IGF Code and Review of the IGC Code and instructed it, taking into account the comments made and decisions taken in plenary, to:

.1 finalize the draft set of amendments to paragraphs 9.6, 9.6.1, 11.6.2, 9.4.7, 12.5, 13.3 and 6.7.3.1.1 of the IGF Code, based on annex 4 to document CCC 8/3;

.2 finalize the draft amendments to paragraph 9.3.1 of the IGF Code, based on annex 6 to document CCC 8/3;

.3 finalize the draft amendments to paragraphs 4.2.2 and 8.4.1 to 8.4.3 of the IGF Code, based on annex 7 to document CCC 8/3;

.4 finalize the draft amendments to part A-1, paragraphs 5.12.1, 6.9.1.1, 9.8.1, 9.8.2, 9.8.4 and to part C-1, paragraph 18.4.1.1.1 of the IGF Code, based on annex 8 to document CCC 8/3;

.5 finalize the draft amendments to paragraphs 9.4.8, 11.3.1 and 15.4.1.3 of the IGF Code, based on annex 3 to document CCC 6/WP.3;

.6 further consider documents CCC 8/3/1, CCC 8/3/2, CCC 8/3/3 and CCC 8/3/4, with a view to preparing amendments to the IGF Code;

.7 further consider document CCC 8/17/2, and advise the Sub-Committee how best to proceed; and

.8 consider whether it is necessary for the correspondence group to be re-established and, if so, prepare terms of reference for consideration by the Sub-Committee.

REPORT OF THE WORKING GROUP ON AMENDMENTS TO THE IGF CODE AND REVIEW OF THE IGC CODE

3.26 Having considered the report of the Working Group on Amendments to the IGF Code and Review of the IGC Code (CCC 8/WP.4), the Sub-Committee approved it in general and took action as described in paragraphs ….

[to be prepared by the Secretariat in consultation with the Chair after the session, based on the Group's report and the actions requested therein, taking into account the decisions taken by the Sub-Committee during subsequent discussions]

4 AMENDMENTS TO THE IGC AND IGF CODES TO INCLUDE HIGH MANGANESE AUSTENITIC STEEL AND RELATED GUIDANCE FOR APPROVING ALTERNATIVE METALLIC MATERIAL FOR CRYOGENIC SERVICE

Background

4.1 The Sub-Committee recalled that MSC 105 approved:

.1 draft amendments to the IGC and IGF Codes on the suitability of high manganese austenitic steel for cryogenic service, with a view to adoption at MSC 106;

.2 MSC.1/Circ.1648 on *Amendments to the Guidelines for the acceptance of alternative metallic materials for cryogenic service in ships carrying liquefied gases in bulk and ships using gases or other low-flashpoint fuels (MSC.1/Circ.1622)*; and

.3 MSC.1/Circ.1599/Rev.2 on *Revised guidelines on the application of high manganese austenitic steel for cryogenic service* (MSC.1/Circ.1599/Rev.1).

4.2 The Sub-Committee also recalled that CCC 7 re-established the Correspondence Group on Suitability of High Manganese Austenitic Steel for Cryogenic Service and instructed it to consider test results to be carried out in accordance with the additional test requirements, in order to evaluate it for qualification of high manganese austenitic steel for ammonia service.

Report of the Correspondence Group

4.3 The Sub-Committee had for its consideration the following documents:

.1 CCC 8/7 (Marshall Islands), reporting on the work of the Correspondence Group, in particular that the Group was unable to complete the work due to the time needed to consider the results of the required testing; and

.2 CCC 8/4/1 (Republic of Korea), providing comments on the report of the Correspondence Group, in particular that there have been technical difficulties in conducting the agreed test procedure and, therefore, the test results have been delayed. The sponsor suggests re-establishing the Correspondence Group to evaluate the test results.

4.4 Having noted the information provided in the above documents, the Sub-Committee noted the following views expressed:

.1 the existing output should be completed, noting the accomplishments under this agenda item; and

.2 any additional work required, e.g. tests and evaluation for qualification of high manganese austenitic steel for ammonia service, could be carried out under agenda item 10 on "Review of the IGC Code".

4.5 Following discussion, the Sub-Committee approved the report of the Correspondence Group in general and instructed the Working Group on Amendments to the IGF Code and Review of the IGC Code to identify any additional work required to complete the output on "Amendments to the IGC and IGF Codes to include high manganese austenitic steel and related guidance for approving alternative metallic material for cryogenic service", noting that the additional work could be carried out under agenda item 10 on "Review of the IGC Code".

Report of the Working Group on Suitability of High Manganese Austenitic Steel

4.6 Having considered the report of the Working Group on Suitability of High Manganese Austenitic Steel (CCC 8/WP.4), the Sub-Committee approved it in general and took action as described in paragraphs 4.7 to 4.[…].

[to be prepared by the Secretariat in consultation with the Chair after the session, based on the Group's report and the actions requested therein, taking into account the decisions taken by the Sub-Committee during subsequent discussions]

5 AMENDMENTS TO THE IMSBC CODE AND SUPPLEMENTS

GENERAL

5.1 The Sub-Committee noted that MSC 105 had adopted amendments (06-21) to the IMSBC Code by resolution MSC.500(105), which is envisaged to enter into force on 1 December 2023 and can be applied on a voluntary basis from 1 January 2023.

5.2 The Sub-Committee also noted that E&T 36 had commenced preparation of the draft amendments (07-23) to the IMSBC Code, to be adopted as a consolidated version.

5.3 The Sub-Committee further noted that, after consideration of the documents submitted to this session, E&T 37 would be instructed to finalize draft amendments (07-23) to the IMSBC Code, for its circulation and subsequent adoption by MSC 107 in June 2023.

REPORT OF E&T 35

5.4 The Sub-Committee considered document CCC 8/5 (Secretariat), the report of E&T 35, and took action as indicated in paragraphs 5.5 to 5.8.

Amendment 06-21 to the IMSBC Code

5.5 The Sub-Committee noted that E&T 35 had finalized the draft amendments (06-21) to the IMSBC Code and had agreed to request the Secretary-General to circulate them in accordance with SOLAS article VIII, for consideration and subsequent adoption by MSC 105 (CCC 8/5, paragraphs 2.1 to 2.23 and annex 1). The Sub-Committee also noted that MSC 105 had adopted amendments (06-21) to the IMSBC Code by resolution MSC.500(105).

MSC.1/Circ.1600

5.6 The Sub-Committee noted that E&T 35 had agreed to the draft revised MSC.1/Circ.1600 and had agreed to request the Secretariat to prepare it for submission to MSC 105 for approval (CCC 8/5, paragraphs 2.5 to 2.6 and annex 2). The Sub-Committee also noted that MSC 105 had approved MSC.1/Circ.1600/Rev.1 on *Guidance for conducting the refined MHB (CR) test*.

Consequential amendments to MSC.1/Circ.1395/Rev.4

5.7 The Sub-Committee noted that E&T 35 had agreed to some consequential amendments to MSC.1/Circ.1395/Rev.4 on *Lists of solid bulk cargoes for which a fixed gas fire-extinguishing system may be exempted or for which a fixed gas fire-extinguishing system is ineffective* and had prepared the draft MSC.1/Circ.1395/Rev.5 for submission to MSC 105 for approval (CCC 8/5, paragraph 2.23 and annex 3). The Sub-Committee also noted that MSC 105 had approved MSC.1/Circ.1395/Rev.5.

Crushed granodiorite

5.8 The Sub-Committee noted the deliberations of E&T 35 on crushed granodiorite (CCC 8/5, paragraphs 3.1 to 3.3).

REPORT OF E&T 36

5.9 The Sub-Committee considered document CCC 8/5/2 (Secretariat), the report of E&T 36, together with the related documents submitted to the session, and took action as indicated in paragraphs 5.10 to 5.26.

Substance identification number for solid bulk cargoes

5.10 The Sub-Committee noted and endorsed the deliberations of E&T 36 on the substance identification number for solid bulk cargoes (CCC 8/5/2, paragraphs 3.2 to 3.6 and annex 2).

5.11 In this context, the Sub-Committee considered document CCC 8/5/4 (Spain), containing information complementary to and in support of the proposal for the identification of solid bulk cargoes in the IMSBC Code, as presented by Spain in document E&T 36/3.

5.12 Following the discussion, the Sub-Committee noted the view expressing concern about the possible implications of introducing substance identification number for IT systems worldwide, administrative burden and consequences for the structure of the IMSBC Code.

5.13 After consideration, the Sub-Committee agreed to refer this matter, along with document CCC 8/5/4, to E&T 37 for further consideration, with a view to providing further advice to CCC 9.

Draft new individual schedules

5.14 The Sub-Committee noted the deliberations of E&T 36 on the draft new individual schedules (CCC 8/5/2, paragraphs 3.7 to 3.44 and annex 1) and agreed to consider the documents submitted to this session based on their relatedness to the draft new individual schedules contained in the report of E&T 36.

Celestine

5.15 The Sub-Committee had for its consideration the following documents submitted by Spain:

.1 CCC 8/5/5, containing a proposal for the inclusion of a new shipping name (CELESTINE CONCENTRATE) in the individual schedule for "Mineral concentrates" in appendix 1 of the IMSBC Code;

.2 CCC 8/5/6, containing a proposal for the inclusion of a new schedule for CELESTINE in appendix 1 of the IMSBC Code;

.3 CCC 8/INF.5, containing the supporting information for the proposal presented in document CCC 8/5/5 to add a new BCSN (CELESTINE CONCENTRATE) to "Mineral concentrates" in appendix 1 of the IMSBC Code; and

.4 CCC 8/INF.6, containing information to support a new individual schedule for CELESTINE in appendix 1 of the IMSBC Code.

5.16 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 the proposals can be supported; however, celestine concentrate should have its own individual schedule;

.2 documents CCC 8/5/6 and CCC 8/INF.6 contain inconclusive information in terms of whether celestine is a group C or a group A cargo and whether it is a cohesive or a non-cohesive cargo; and

.3 the proposal in document CCC 8/5/6 can be supported; however, there are some inconsistencies in the draft new individual schedule concerning bulk density and stowage factor, and weather precautions.

5.17 After consideration, the Sub-Committee agreed to the proposals, in principle, and decided to refer these documents to E&T 37 for consideration and incorporation, as appropriate, into draft amendment 07-23 to the IMSBC Code.

Crushed granodiorite

5.18 The Sub-Committee noted that E&T 36 had agreed to the draft new individual schedule for crushed granodiorite (CCC 8/5/2, paragraphs 3.17 to 3.22 and annex 1).

5.19 In this context, the Sub-Committee had for its consideration the following documents submitted by Norway:

.1 CCC 8/5/9, containing a proposal to amend the draft new group C individual schedule agreed at E&T 36 for crushed granodiorite to group A, as additional testing has shown that the finest fraction of the cargo has the potential to liquefy; and

.2 CCC 8/INF.12, containing information to support updating the draft new individual schedule for crushed granodiorite agreed at E&T 36 from group C to group A.

5.20 The Sub-Committee noted the comments by Norway that the BCSN for the draft new individual schedule should be CRUSHED GRANODIORITE FINES and that grain size for this cargo should be up to 22 mm.

5.21 After consideration, the Sub-Committee agreed to the proposals, in principle, and taking into consideration the comments made, decided to refer these documents to E&T 37 for consideration and incorporation, as appropriate, into draft amendment 07-23 to the IMSBC Code.

Ground granulated blast furnace slag powder

5.22 The Sub-Committee had for its consideration document CCC 8/5/17 (China), providing comments on paragraphs 3.32 to 3.35 of document CCC 8/5/2 (Secretariat) and inviting the Sub-Committee to consider the proposal in document E&T 36/3/6 (China) to include ground granulated blast furnace slag powder as group A cargo in appendix 1 of the IMSBC Code.

5.23 After consideration, the Sub-Committee agreed to the proposal, in principle, and decided to refer these documents to E&T 37 for consideration and incorporation, as appropriate, into draft amendment 07-23 to the IMSBC Code.

Safe transport of fumigated cargo

5.24 The Sub-Committee noted the deliberations of E&T 36 on safe transport of fumigated cargo (CCC 8/5/2, paragraphs 3.45 to 3.47).

Minor corrections

5.25 The Sub-Committee noted the deliberations of E&T 36 on minor corrections to the IMSBC Code (CCC 8/5/2, paragraph 3.48 and annex 1).

Draft amendments (07-23) to the IMSBC Code

5.26 The Sub-Committee agreed to the draft amendment 07-23 to the IMSBC Code, as prepared by E&T 36 (CCC 8/5/2, paragraph 3.49 and annex 1), with a view to finalization by E&T 37.

PROPOSALS FOR AMENDMENTS TO THE EXISTING INDIVIDUAL SCHEDULES AND PROVISIONS IN THE IMSBC CODE AND RELATED INSTRUMENTS

Alignment of the IMSBC Code with SOLAS on declaration of solid bulk density

5.27 The Sub-Committee had for its consideration document CCC 8/5/1 (Liberia et al.), proposing a minor correction of the IMSBC Code in order to reflect an existing requirement for shippers to declare a technical aspect of cargoes which is in force through SOLAS regulation XII/10 (solid bulk cargo density declaration); and also proposing to bring the issue to the attention of stakeholders by means of an MSC circular, pending the voluntary application of the amendments to the IMSBC Code.

5.28 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 the proposals can be supported; in addition, it should be explored whether there are any other inconsistencies between the SOLAS Convention and the IMSBC Code; and

.2 the proposals can otherwise be supported, but replication of SOLAS regulations XII/1, XII/2, XII/6 and XII/10 in provision 1.6 of the IMSBC Code, as proposed in paragraph 7.2 of document CCC 8/5/1, cannot be supported.

5.29 After consideration, apart from the proposal in paragraph 7.2 of document CCC 8/5/1, the Sub-Committee, agreed, in principle, to the proposal for amendment to the IMSBC Code, and decided to refer the document to E&T 37 for consideration and incorporation, as appropriate, into draft amendment 07-23 to the IMSBC Code. With regard to the draft MSC circular, the Sub-Committee agreed, in principle, to the proposal, and decided to refer the issue to E&T 37 for further consideration, with a view to submission to MSC 107 for approval.

Amendments to MSC.1/Circ.1453/Rev.1 and MSC.1/Circ.1454/Rev.1

5.30 The Sub-Committee had for its consideration document CCC 8/5/7 (France), proposing amendments to MSC.1/Circ.1453/Rev.1 and MSC.1/Circ.1454/Rev.1, stemming from amendments 06-21 to the IMSBC Code, which introduced definitions for "dynamic separation" and "cargoes which may undergo dynamic separation", affecting various sections of the IMSBC Code, including section 7.

5.31 After consideration, the Sub-Committee agreed to the proposals, in principle, and decided to refer the document to E&T 37 for further consideration, with a view to submission to MSC 107 for approval.

The IMSBC Code schedules for iron ore pellets

5.32 The Sub-Committee had for its consideration document CCC 8/5/8 (Brazil et al.), which:

.1 noted that the IMSBC Code contains three individual schedules which could be used for shipment of iron ore pellets;

.2 based on consultations with the iron ore industry, proposed that the IRON ORE PELLETS schedule is the correct one and that the TACONITE PELLETS schedule should be deleted from the Code; and

.3 noted that the application of the third schedule, PELLETS (concentrates), is unclear and it should not be used for iron ore pellets; and proposed that it should, nevertheless, remain in the Code.

5.33 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 the individual schedule for TACONITE PELLETS is still in use and should be retained;

.2 any proposal for deleting an existing individual schedule should be based on scientific evidence; and

.3 there is no provision in section 1 of the IMSBC Code expressly mandating the deletion of an individual schedule based on its non-use.

5.34 In this context, the Sub-Committee agreed to refer this issue to E&T 37 for further consideration, if time permitted, in order to provide advice to the co-sponsors of document CCC 8/5/8.

Proposed amendments to the individual schedule for COAL

5.35 The Sub-Committee had for its consideration documents CCC 8/5/10 and CCC 8/INF.15 (Australia), proposing an amendment to the individual schedule for COAL to clarify the classification and carriage requirements for coal returning a positive result to the N.4 self-heating test (exempted if transported in packages of not more than 450 L volume or exempted if transported in packages of not more than 3 m^3 volume); and containing the report of the Australian Coal Industry's investigation of self-heating test methods for coal.

5.36 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 even though the N.4 test may not be completely representative for COAL, classification of COAL should continue to be based on this test, which is accepted worldwide, instead of research; an opposite decision to this would create a precedent;

.2 if the N.4 test is unreliable based on research, this information should be brought to the UN level for further consideration;

.3 the N.4 test should be removed from application to COAL, and all COAL and CHARCOAL cargoes should be classified as 4.2;

.4 the proposal would reduce safety; and

.5 while the proposal needs further consideration, preparation of a circular could be a way forward.

5.37 After consideration, the Sub-Committee agreed to refer this matter to E&T 37 for further consideration, and to provide further advice to CCC 9.

Enclosed space fatalities aboard ships during the period 2000 to 2021

5.38 The Sub-Committee had for its consideration document CCC 8/5/11 (IBTA), providing a summary of research carried out to identify and analyse all reported enclosed space-related fatalities across all ship types for the period 2000 to 2021, and aiming to place previous data regarding enclosed space fatalities during the carriage of solid bulk cargoes in context in relation to the total number of enclosed space fatalities and to support proposals to amend relevant provisions of both the IMSBC Code and the *Revised recommendations for entering enclosed spaces aboard ships* (resolution A.1050(27)).

5.39 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 while the scope of existing agenda item 8 on "Revision of the Revised recommendations for entering enclosed spaces aboard ships (resolution A.1050(27))" is limited, document MSC 106/16/1 (China), proposing a new output to revise resolution A.1050(27), should be taken into account in this regard; and

.2 it should be noted that CCC 6 (CCC 6/14, paragraph 5.26) had agreed that amending section 3 of the IMSBC Code, as proposed in paragraph 12.1 of document CCC 6/5/6 (United Kingdom et al.), would require a submission to MSC for a new output.

5.40 After consideration, the Sub-Committee agreed that there were no specific actions taken under agenda item 5 during this session and that the relevant part of document CCC 8/5/11 regarding amendments to A.1050(27) could be taken into account by a future session, if a new output to revise resolution A.1050(27) was agreed by the Maritime Safety Committee (see also paragraph 8.[…]).

Transport of fish meal in bulk

5.41 The Sub-Committee had for its consideration document CCC 8/5/12 (Germany), proposing an alignment of the stabilization requirements for fish meal in the IMSBC Code with the respective requirements in the IMDG Code and to classify fish meal in bulk as MHB (SH) instead of class 9.

5.42 After consideration, the Sub-Committee agreed to the proposal, in principle, and decided to refer the document to E&T 37 for consideration and incorporation, as appropriate, into draft amendment 07-23 to the IMSBC Code.

Proposal for an amendment to the IMSBC Code to clarify the carriage requirement of spare charges for SCBAs (self-contained breathing apparatuses) required to be provided on board solely by the IMSBC Code and the proposed draft unified interpretation of SOLAS regulation II-2/19.3.6.2 and the IMSBC Code

5.43 The Sub-Committee had for its consideration the following documents submitted by IACS:

.1 CCC 8/5/13, proposing to amend the IMSBC Code to clarify the carriage requirement of spare charges for SCBAs (self-contained breathing apparatuses) required solely by the IMSBC Code; and

.2 CCC 8/12/4, proposing a draft unified interpretation of the requirements for additional self-contained breathing apparatus contained in SOLAS chapter II-2 and the IMSBC Code.

5.44 Following the discussion, the Sub-Committee noted the following views expressed on document CCC 8/5/13:

.1 when SOLAS regulation II-2/19.3.6.2 was introduced, the IMSBC Code should have been amended in order to remove the requirements for SCBAs in addition to those required by SOLAS regulation II-2/10.10;

.2 SOLAS regulation II-2/19 specifies the requirements in terms of the number of SCBAs for ships to be had on board;

.3 the proposal would need a new output; and

.4 the proposal can be supported.

5.45 With regard to document CCC 8/12/4, the Sub-Committee noted the following views:

.1 the proposal can be supported; and

.2 as there is a link to document CCC 8/5/13, it can be noted in a similar vein that when SOLAS regulation II-2/19.3.6.2 was introduced, the IMSBC Code should have been amended in order to remove the requirements for SCBAs in addition to those required by SOLAS regulation II-2/10.10; and it can also be noted that deleting the requirements from the individual schedules would not need a new output.

5.46 After consideration, the Sub-Committee agreed to refer documents CCC 8/5/13 and CCC 8/12/4 to E&T 37 for further consideration, and advice to CCC 9 accordingly.

Proposal to amend the existing individual schedule for Wood products – General

5.47 The Sub-Committee had for its consideration document CCC 8/5/15 (China), proposing amendments to the existing individual schedule for Wood products – General, in order to clarify the provisions on transportation of wood products.

5.48 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 oxygen depletion has been discussed at length previously, and it could be noted that E&T 29 had agreed that the criteria for determination of "Other hazards (OH)" is necessary and had invited the interested parties to submit

proposals to CCC 5 in this regard (CCC 5/5, paragraph 3.8); however, in the absence of concrete proposals for these criteria, it could be argued that it is not possible to have further discussion on this issue for the time being;

.2 a proposal to amend section 9 of the IMSBC Code would require a new output; and

.3 similar existing individual schedules should be taken into account.

5.49 In this context, the Sub-Committee agreed to invite the delegation of China to take into account the comments made above when considering how to proceed on this matter in future.

PROPOSALS FOR NEW INDIVIDUAL SCHEDULES

Chemical gypsum powder

5.50 The Sub-Committee had for its consideration documents CCC 8/5/3 and CCC 8/INF.4 (Finland), proposing a new individual schedule for chemical gypsum powder for inclusion in the IMSBC Code and containing the cargo information to support the proposal for a new individual schedule.

5.51 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 instead of introducing a new individual schedule, the existing schedule for CHEMICAL GYPSUM should be modified;

.2 a new individual schedule could be introduced, while the option of modifying the existing schedule should be avoided because the chemical formulas of these substances are different; and

.3 proposals for modifying the existing schedule should be invited.

5.52 After consideration, the Sub-Committee agreed to refer documents CCC 8/5/3 and CCC 8/INF.4 to E&T 37 for further consideration, if time permitted, with a view to providing further advice to CCC 9, taking into account the comments made above.

Contaminated soil

5.53 The Sub-Committee had for its consideration documents CCC 8/5/14 and CCC 8/INF.18 (Ireland), proposing a new individual schedule for contaminated soil for inclusion in the IMSBC Code and containing the cargo information to support the proposed new individual schedule.

5.54 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 this is a complex substance and its characteristics may vary considerably; hence, introducing one individual schedule may not solve the issue;

.2 there is no evidence on whether this waste material is hazardous or not, as no waste characterization analysis has been provided, as required by 10.6 of the IMSBC Code;

.3 the presence of dangerous contaminants such as heavy metals and hydrocarbons can lead to long-term toxic effects for people, as well as potentially generating the accumulation of harmful or flammable vapours inside the cargo holds during transport;

.4 the proposed generic schedule can potentially allow materials to be transported with inadequately assessed risks;

.5 in the proposed schedule, the material is classified as group A, but the emergency procedures, which are only applicable to group B materials, are also indicated;

.6 according to 4.1.2 of the IMSBC Code, when waste is being transported for disposal, the BCSN must be preceded by the word "WASTE";

.7 the sentence in the section for "Loading" is only applicable for cargoes with a stowage factor less than 0.56 m^3/t; and

.8 wastes are frequently carried under UN 3077 in packaged form; and in this regard, it could be noted that the UN number is not easily added to a schedule in the IMSBC Code.

5.55 After consideration, the Sub-Committee agreed to refer documents CCC 8/5/14 and CCC 8/INF.18 to E&T 37 for further consideration, if time permitted, with a view to providing further advice to CCC 9, taking into account the comments made above.

Magnesite fines

5.56 The Sub-Committee had for its consideration documents CCC 8/5/16 and CCC 8/INF.20 (China), proposing to include a new individual schedule for magnesite fines in the IMSBC Code and containing the cargo information to support the proposed new individual schedule.

5.57 After consideration, the Sub-Committee agreed to the proposal, in principle, and decided to refer these documents to E&T 37 for consideration and incorporation, as appropriate, into draft amendment 07-23 to the IMSBC Code.

DRAFT AMENDMENT 07-23 TO THE IMSBC CODE AND INSTRUCTIONS TO THE E&T GROUP

5.58 Having considered the above matters, the Sub-Committee authorized E&T 37 to finalize draft amendment 07-23 to the IMSBC Code, based on the documents submitted to CCC 8 and taking into account comments made and decisions taken by the Sub-Committee, with a view to submitting the draft consolidated edition of the IMSBC Code, incorporating draft amendment 07-23, to MSC 107 for consideration and adoption, and to submit a written report to CCC 9.

5.59 Subsequently, the Sub-Committee requested the Secretary-General to circulate draft amendment 07-23 to the IMSBC Code, to be finalized by E&T 37, in accordance with SOLAS article VIII, for consideration and subsequent adoption by MSC 107.

5.60 The Sub-Committee noted that the provisional agenda for E&T 37 was available in document E&T 37/1.

6 AMENDMENTS TO THE IMDG CODE AND SUPPLEMENTS

GENERAL

6.1 The Sub-Committee recalled that MSC 105 had adopted amendments to the IMDG Code (41-22) by resolution MSC.501(105), which is envisaged to enter into force on 1 January 2024 and can be applied on a voluntary basis from 1 January 2023.

6.2 The Sub-Committee also recalled that CCC 7 had re-established the Correspondence Group on a Review of Maritime Special Provisions and had instructed the Group to report to this session.

6.3 The Sub-Committee noted that, after consideration of the submissions under this agenda item, it should provide clear advice, instruction and authorization to E&T 38, to be convened in the spring of 2023, in order to prepare the draft text of amendment 42-24 to the IMDG Code, with a view to adoption at MSC 108 in 2024.

REPORT OF E&T 35

6.4 The Sub-Committee considered the report of E&T 35 (CCC 8/6), together with the related documents submitted to the session and, having approved it in general, took action as indicated in paragraphs 6.5 to 6.18.

Editorial corrections to the English version of amendment 40-20 to the IMDG Code

6.5 The Sub-Committee noted that E&T 35 had finalized the editorial corrections applicable to the English version of amendment 40-20 to the IMDG Code (resolution MSC.477(102)) and had agreed to request the Secretariat to issue a corrigendum before 1 June 2022, the date when amendment 40-20 entered into force (CCC 8/6, paragraph 2.1 and annex 1). The Secretariat had prepared and issued the corrigendum, as set out in document MSC 102/24/Add.2/Corr.1.

Editorial corrections to the French and Spanish versions of amendment 40-20 to the IMDG Code

6.6 The Sub-Committee noted that the Group had finalized the editorial corrections applicable to the French version of amendment 40-20 to the IMDG Code (resolution MSC.477(102)) and had agreed to request the Secretariat to issue separate corrigenda, containing the editorial corrections (CCC 8/6, paragraphs 2.2 to 2.3 and annex 2).

The Secretariat had prepared and issued the corrigenda, as set out in document MSC 102/24/Add.2/Corr.1, for the French and Spanish versions.

Outdated references to ISO standards

6.7 The Sub-Committee noted the deliberations of E&T 35 on correction of outdated references to ISO standards (CCC 8/6, paragraphs 2.8 to 2.10).

"Diacetyl, *see*" and "2-Phenylpropene, *see*" in the index of the IMDG Code

6.8 As invited by E&T 35, the Sub-Committee agreed to invite interested Member States and international organizations to provide more relevant information on "Diacetyl, *see*" and "2-Phenylpropene, *see*", as contained in the index of the IMDG Code (CCC 8/6, paragraphs 2.11 to 2.14).

Provisions concerning transport operations in the IMDG Code

6.9 The Sub-Committee noted the deliberations of E&T 35 on the proposed amendment to provisions concerning transport operations in the IMDG Code (CCC 8/6, paragraphs 3.4 to 3.11) and agreed to invite interested Member States and international organizations to submit further proposals.

Output on fibre-reinforced plastics

6.10 The Sub-Committee noted that E&T 35 had agreed to inform MSC 105 on the completion of the work on the output on "Amendments to the IMDG Code related to portable tanks with shells made of fibre-reinforced plastics (FRPs) for multimodal transportation of dangerous goods" (CCC 8/6, paragraph 3.16). Consequently, MSC 105 had agreed to delete the output from the post-biennial agenda of the Committee, as the work under this output had already been covered by amendment 41-22 of the IMDG Code (MSC 105/20, paragraph 18.49).

Further advice from IAEA

6.11 The Sub-Committee noted the deliberations of E&T 35 on proposed amendments emanating from IAEA (CCC 8/6, paragraphs 3.17 to 3.19) and agreed to request the Secretariat to seek further advice from IAEA.

Informing UNSCETDG

6.12 The Sub-Committee noted that the Secretariat had submitted document UN/SCETDG/59/INF.29 to UNSCETDG 59, informing of the outcome of E&T 35.

Consequently, the Secretariat had submitted document ST/SG/AC.10/C.3/2022/31 to UNSCETDG 60, proposing corresponding amendments to the UN Model Regulations in order to align the UN Model Regulations with the IMDG Code. These amendments had been adopted at UNSCETDG 60 (ST/SG/AC.10/C.3/120, paragraph 90).

Container data loggers and tracking devices

6.13 The Sub-Committee noted the deliberations of E&T 35 on container data loggers and tracking devices and the invitation to interested Member States and international organizations to continue working together on this issue, with a view to submitting further proposals (CCC 8/6, paragraphs 3.23 to 3.26).

6.14 In this context, the Sub-Committee had for its consideration document CCC 8/6/4 (United States et al.), containing proposed criteria for devices in use or intended for use during transport, such as data loggers, sensors and cargo trackers that contain dangerous goods that are attached to or placed in packages, overpacks, containers, cargo transport units or load compartments.

6.15 After consideration, the Sub-Committee agreed to the proposal in document CCC 8/6/4, in principle, and referred the document to the Drafting Group on Amendments to the IMDG Code, for consideration and incorporation, as appropriate, into draft amendment 42- 24 to the IMDG Code.

Amendments (41-22) to the IMDG Code

6.16 The Sub-Committee noted that E&T 35 had finalized the draft amendments (41-22) to the IMDG Code and had agreed to request the Secretariat to circulate them in accordance with SOLAS article VIII, for consideration and subsequent adoption by MSC 105 (CCC 8/6, paragraph 3.32 and annex 3). MSC 105 had adopted the amendments by resolution MSC.501(105).

Amendments to the EmS Guide

6.17 The Sub-Committee noted that E&T 35 had finalized the consequential amendments to the EmS Guide and had agreed to request the Secretariat to submit them to MSC 105 for approval (CCC 8/6, paragraphs 4.1 to 4.2 and annex 4). MSC 105 had approved MSC.1/Circ.1588/Rev.2 on the *Revised emergency response procedures for ships carrying dangerous goods* (EmS Guide), containing a revised consolidated version of the Guide.

Editorial amendments to MSC.1/Circ.1361

6.18 The Sub-Committee noted that E&T 35 had finalized the editorial amendments to MSC.1/Circ.1361 and had agreed to request the Secretariat to submit them to MSC 105 for approval (CCC 8/6, paragraphs 4.3 to 4.6 and annex 5). MSC 105 had approved MSC.1/Circ.1361/Rev.1 on the *Revised recommendations on the safe use of pesticides in ships applicable to the fumigation of cargo transport units*.

REPORT OF THE CORRESPONDENCE GROUP ON A REVIEW OF MARITIME SPECIAL PROVISIONS

6.19 The Sub-Committee recalled that CCC 7 had re-established the Correspondence Group on a Review of Maritime Special Provisions, with the terms of reference set out in paragraph 6.48 of document CCC 7/15 (the report of CCC 7).

6.20 In this context, the Sub-Committee had for its consideration the following documents submitted to this session:

.1 CCC 8/6/1 (Germany), containing the report of the Correspondence Group on a Review of Maritime Special Provisions;

.2 CCC 8/6/16 (Japan), providing comments on the report of the Correspondence Group with regard to the consequential amendment to the IMSBC Code, in case the Sub-Committee decides to treat CHARCOAL as dangerous goods regardless of the result of the UN N.4 test;

.3 CCC 8/6/17 (CEFIC), containing a reaction to the report of the Correspondence Group (CCC 8/6/1) relating to amendments to the IMDG Code for the transport of UN 1361 CARBON (animal or vegetable) and the implications for UN 1362, CARBON, ACTIVATED; and proposing amendments and alternative solutions to the report of the Correspondence Group; and

.4 CCC 8/INF.16 (CEFIC), presenting CEFIC's position on amendments to the IMDG Code related to the reclassification and transport of UN 1361 CARBON (animal or vegetable) and the implications for UN 1362, CARBON, ACTIVATED.

6.21 With regard to the report of the Correspondence Group, as set out in document CCC 8/6/1, the Sub-Committee took action, as set out in the following paragraphs.

Applicability of the UN N.4 test for consignments of charcoal and the related special provisions

6.22 The Sub-Committee noted the Group's discussion on the applicability of the UN N.4 test for consignments of charcoal in paragraphs 5 to 10 of the report and the invitation to delete special provisions 223 and 925 in the entry for UN 1361 in the Dangerous Goods List (CCC 8/6/1, paragraph 25.1).

6.23 Following the discussion, the Sub-Committee noted the following views on this matter:

.1 the proposals in paragraphs 25 and 26 of document CCC 8/6/1 should be limited to bulk shipments, as their application to smaller shipments is not justified; as an alternative, exemptions and a transition period should be applied to smaller shipments; and

.2 all related documents should be referred to the Drafting Group on Amendments to the IMDG Code for further consideration.

6.24 In this context, the Sub-Committee instructed the Drafting Group on Amendments to the IMDG Code to further develop amendments to the IMDG Code concerning applicability of the UN N.4 test for consignments of charcoal and the related special provisions, taking into account comments made above and documents CCC 8/6/1, CCC 8/6/17 and CCC 8/INF.16.

Ambient temperature

6.25 The Sub-Committee noted the Group's discussion on the term "ambient temperature" in paragraph 11 of the report and the invitation to include the proposed definition in 1.2.1 of the IMDG Code (CCC 8/6/1, paragraph 24).

6.26 After consideration, the Sub-Committee instructed the Drafting Group to further develop amendments to the IMDG Code concerning ambient temperature, taking into account documents CCC 8/6/1, CCC 8/6/17 and CCC 8/INF.16.

Stowage of UN 1361

6.27 The Sub-Committee noted the Group's discussion on stowage requirements for UN 1361 in paragraphs 19 and 20 of the report and the consequence that the proposed

amendment of the stowage category from A to C would prohibit the transport of UN 1361 on ferry ships without on-deck stowage locations and also as break-bulk in the cargo hold of a general cargo ship; and if these consequences are deemed acceptable, the invitation to approve the assignment of stowage category C to UN 1361 (CCC 8/6/1, paragraph 25.2).

6.28 After consideration, the Sub-Committee agreed not to approve the proposed amendment of stowage category from A to C to UN 1361.

Stowage in accessible position

6.29 The Sub-Committee noted the Group's discussion on a requirement of "stowage in accessible position" in paragraph 21 of the report and noted that no concrete proposal had been agreed by the Group.

Special provisions for UN 1361

6.30 The Sub-Committee noted the Group's invitation to approve the new special provision 9xa for UN 1361, as a consequence of the deletion of special provisions 223 and 925 in the entry for UN 1361 (CCC 8/6/1, paragraph 26.1).

6.31 Following the discussion, the Sub-Committee noted the following views on this matter:

.1 the exemption for non-activated carbon blacks of mineral origin should be retained in the new special provision, as proposed in documents CCC 8/6/17 and CCC 8/INF.16; and

.2 the proposal for a new special provision 9xa is not adequate and needs further consideration.

6.32 In this context, the Sub-Committee instructed the Drafting Group to further develop amendments to the IMDG Code concerning the special provisions for UN 1361, taking into account comments made above and documents CCC 8/6/1, CCC 8/6/17 and CCC 8/INF.16.

Special provisions for UN 1362

6.33 The Sub-Committee noted the Group's considerations related to activated carbon (UN 1362) in paragraph 22 of the report and noted the proposed options for the new special provision 9xb, replacing or revising UN 925 in the entry for UN 1362; and the invitation to decide which option is to be taken (CCC 8/6/1, paragraph 26.2).

6.34 After consideration, and considering that the majority of those who spoke in plenary supported option 1, the Sub-Committee instructed the Drafting Group to further develop amendments to the IMDG Code concerning the special provisions for UN 1362, taking into account documents CCC 8/6/1, CCC 8/6/17 and CCC 8/INF.16.

Vanning certificate

6.35 The Sub-Committee noted the Group's discussion on the vanning certificate in paragraph 16 of the report and the invitation to decide whether such a certificate shall be required for UN 1361 and, if so decided, the invitation to approve the text of the proposed special provision 9xc (CCC 8/6/1, paragraph 26.3).

6.36 Following the discussion, the Sub-Committee noted the following views on this matter:

.1 the proposed requirement for a vanning certificate would cause practical difficulties; and such a requirement should in any case be based on scientific evidence;

.2 vanning certificate is already in use for many shipments of charcoal; and

.3 the proposal for a vanning certificate can be supported but needs further consideration.

6.37 After consideration, the Sub-Committee agreed to invite interested Member States and international organizations to submit further proposals on this matter to E&T 38.

Packing provisions for UN 1361

6.38 The Sub-Committee noted the Group's discussion on the packing provisions in paragraph 17 of the report and the invitation to approve the deletion of UN 1361 under PP12 (CCC 8/6/1, paragraph 27).

6.39 After consideration, the Sub-Committee agreed to the deletion of UN 1361 under PP12, in principle, and decided to refer this matter to E&T 38 for consideration and incorporation, as appropriate, into draft amendment 42-24 to the IMDG Code.

Documentation requirements for UN 1361

6.40 The Sub-Committee noted the Group's discussion on the documentation requirements in paragraph 14 of the report and the invitation to approve inclusion of the new provision 5.4.1.5.18 (CCC 8/6/1, paragraph 28).

6.41 After consideration, the Sub-Committee instructed the Drafting Group to further develop amendments to the IMDG Code concerning the documentation requirements for UN 1361.

Finalization of the approved amendments

6.42 The Sub-Committee noted the Group's invitation for the E&T Group to finalize the above-mentioned amendments.

Possible consequences for the IMSBC Code

6.43 The Sub-Committee noted the Group's deliberations regarding possible consequences for the IMSBC Code (CCC 8/6/1, paragraph 23).

6.44 After consideration, the Sub-Committee instructed the Drafting Group to further consider possible consequences for the IMSBC Code, taking into account documents CCC 8/6/1 and CCC 8/6/16.

Basic documentation requirements for goods not subject to the IMDG Code

6.45 The Sub-Committee noted the Group's discussion in paragraph 35 of the report on the different views on the issue whether goods not subject to the IMDG Code should be subject to basic documentation requirements of the IMDG Code. The Sub-Committee further noted the Group's invitation to consider the options provided in paragraph 36 of the report and the allocation of these options to the various maritime special provisions in the annex to the report; and the invitation to decide which option shall be taken, or when deemed appropriate, the invitation to establish a working group at this session in order to prepare the decision of the Sub-Committee.

6.46 Following the discussion, the Sub-Committee noted the following views on this matter:

.1 option 1 can be supported; and the annex to document CCC 8/6/1 could be the basis for further work;

.2 option 2 can be supported, in principle; however, decisions on retaining individual maritime special provisions should be made on a case-by-case basis; and

.3 option 3 can be supported.

6.47 After consideration, the Sub-Committee agreed to invite interested Member States and international organizations to submit further proposals on this matter to E&T 38.

Special provisions for seed cake

6.48 The Sub-Committee noted the Group's remarks in paragraph 38 of the report that the special provisions 142, 388 and 929 in the IMDG Code are inconsistent with the special provisions 142, 388 and the criteria for the assignment of UN 2217 in the UN Model Regulations.

6.49 After consideration, the Sub-Committee agreed to refer this matter to E&T 38 for further consideration.

Competent authorities

6.50 The Sub-Committee noted the Group's invitation to consider the options related to the question of competent authorities in paragraphs 40 to 43 of the report; and to decide which option shall be taken, and in case a working group is established, based upon the result provided by that working group.

6.51 After consideration, the Sub-Committee agreed to:

.1 option 2 concerning SP907 and SP926, along with deletion of the words "by a body appointed by the shipper to act as competent authority"; and

.2 option 1 concerning SP925 and SP964, along with deletion of the words "by a body appointed by the shipper to act as competent authority".

6.52 In this context, the Sub-Committee instructed the Drafting Group to further consider this matter, taking into account the decisions made above.

Amendments to 5.4.4 of the IMDG Code

6.53 The Sub-Committee noted the Group's invitation to consider the options in paragraphs 47 and 48 of the report for the improved text of 5.4.4 of the IMDG Code; and decide which option shall be taken, and in case a working group is established, based upon the result provided by that working group.

6.54 After consideration, the Sub-Committee agreed to refer this matter to E&T 38 for further consideration.

Approved amendments and outstanding issues from document CCC 7/6/2

6.55 The Sub-Committee noted the Group's invitation to instruct the E&T Group to finalize approved amendments concerning basic documentation requirements for goods not subject to the IMDG Code, competent authorities and provision 5.4.4 of the IMDG Code; and to consider the outstanding issues from paragraph 41 of document CCC 7/6/2 (Germany), and report the outcome to CCC 9.

6.56 Taking into account the decisions made above, the Sub-Committee agreed to refer the outstanding issues from paragraph 41 of document CCC 7/6/2 to E&T 38 for further consideration.

PROPOSALS RELATED TO AMENDMENT 42-24

Amendments to the provisions of the IMDG Code to clarify the application of paragraph 2.10.2.7 thereof

6.57 The Sub-Committee had for its consideration document CCC 8/6/2 (France), proposing amendments to the provisions of the IMDG Code to clarify the application of paragraph 2.10.2.7 thereof by limiting the exemption from the provisions of the IMDG Code (in particular, the provisions relating to the marking and labelling of packages, placarding and marking of transport units and the transport document) to the "marine pollutant" characteristic alone.

6.58 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 the proposal can be supported, in general; however, chapter 5.4 of the IMDG Code should be amended so as to ensure that these substances are listed in dangerous goods transport document;

.2 the proposal can be supported but requires some editorial modifications;

.3 this issue can be solved by adding a clarifying text to the end of paragraph 2.10.2.7;

.4 amending chapter 5.4 of the IMDG Code should be avoided; and

.5 it should be noted that also SP969 applies to these substances (see also annex 2 of document CCC 7/6/2).

6.59 After consideration, the Sub-Committee agreed to the proposal, in principle, and decided to refer the document to E&T 38 for consideration and incorporation, as appropriate, into draft amendment 42-24 to the IMDG Code, taking into account SP969.

Amendment to paragraph 7.3.3.14 of the IMDG Code

6.60 The Sub-Committee had for its consideration document CCC 8/6/3 (France), proposing to amend paragraph 7.3.3.14 of the IMDG Code as, according to France, the current wording of 7.3.3.14 of the IMDG Code, in which a reference to the CTU Code is included in the body text of the Code, could wrongly lead one to believe that the CTU Code is a mandatory instrument, whereas it is only a recommended instrument, published by means of a circular (MSC.1/Circ.1497).

6.61 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 the wording of 7.3.3.14 of the IMDG Code should be retained;

.2 the proposal can be supported; and

.3 adding a footnote would be consistent with the way footnotes are currently used in the IMDG Code.

6.62 After consideration, the Sub-Committee agreed to the proposal, in principle, and decided to refer the document to E&T 38 for consideration and incorporation, as appropriate, into draft amendment 42-24 to the IMDG Code.

Mixed stowage of articles and substances of compatibility groups C, D and E

6.63 The Sub-Committee had for its consideration document CCC 8/6/5 (United States), containing a proposal for amendment to the existing requirements for mixed stowage of articles and substances of compatibility groups C, D and E. The document identifies a potential mixed load of explosive articles and substances that the IMDG Code does not currently address, and proposes a modification of mixed stowage requirements.

6.64 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 the proposal can be supported;

.2 similar amendments are also needed for Note 1 and Note 4;

.3 there is also a need to request UNTDG to clarify the meaning of terms "goods", "substances" and "articles"; and

.4 the proposal cannot be supported because compatibility group E refers to articles; in addition, this matter should be considered by UNTDG first.

6.65 After consideration, the Sub-Committee agreed to invite the delegation of the United States to consider submitting a proposal on this matter to UNTDG.

Special provisions for the transport of vehicles

6.66 The Sub-Committee had for its consideration document CCC 8/6/6 (United States), containing a proposal to replace special provisions 961 and 962 with a new special provision 9XX in response to recent fire incidents occurring on vehicle carriers.

6.67 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 the proposal could be a basis for future work;

.2 it should be noted that the IMDG Code should not include requirements for administrations;

.3 reasoning regarding the limitation of fuel volume in conventional vehicles cannot be supported; the proposal in paragraphs 13 and 1.11 of the annex to document CCC 8/6/6, specifying that a damaged battery-powered vehicle should be able to be transported with the battery still in the vehicle in a separate CTU is problematic and inappropriate; it should also be noted that MSC 105 agreed to include in its post-biennial agenda an output on "Evaluation of adequacy of fire protection, detection and extinction arrangements in vehicle, special category and ro-ro spaces in order to reduce the fire risk of ships carrying new energy vehicles", assigning the SSE Sub-Committee as the associated organ; in addition, it may be appropriate to distinguish the requirements between ro-ro cargo ships, pure car carriers and ro-ro passenger ships;

.4 the proposal raises concerns regarding administrative burden, quality of inspections and state of charge; in addition, document CCC 8/6/10 (Germany) should be taken into account in future work;

.5 the proposal is problematic for ro-ro cargo ships and ro-ro passenger ships;

.6 incidents with used and damaged vehicles should be considered; however, applying the requirements to new vehicles is not necessary;

.7 the future work should be science-based; and

.8 a correspondence group could be established to progress this work.

6.68 After consideration, the Sub-Committee instructed the Drafting Group to prepare draft terms of reference for a correspondence group, with a view to submitting a report to CCC 9.

Proposed amendments to provision 7.6.2.8.4

6.69 The Sub-Committee had for its consideration documents CCC 8/6/7 and CCC 8/INF.14 (ICHCA), proposing a revision of provision 7.6.2.8.4 of the IMDG Code to clarify arrangements for emergency access to ammonium nitrate when stowed under deck; and providing the supporting information to the proposed revision of provision 7.6.2.8.4.

6.70 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 The proposal can be supported in general. However, with regard to the ability to open hatches, further scientific evidence for the figure "30%" is required.

.2 It is not certain whether the addition of text regarding the role of the document of compliance for dangerous goods is helpful. The proposal to amend the provision to make it clear that "hatch covers" is intended to include hatches, pontoons and tween decks between the main deck hatch covers and the ammonium nitrate cargo, can be supported. The requirement is that the cargo is carried in a clean cargo space that is capable of being opened in an emergency, and that applies to every cargo space below the weather deck hatch covers where the cargo is stowed. The wording concerning the need to open hatches can easily be read to mean that only the weather deck hatch covers need to be able to be opened.

.3 The proposal would improve safety therefore should be supported.

6.71 After consideration, the Sub-Committee agreed to the proposal, in principle, and decided to refer these documents to E&T 38 for consideration and incorporation, as appropriate, into draft amendment 42-24 to the IMDG Code.

Limited quantity provisions for UN 2956 MUSK XYLENE

6.72 The Sub-Committee had for its consideration document CCC 8/6/8 (Germany), addressing differences in the provisions for the transport of UN 2956 MUSK XYLENE in limited quantities between the IMDG Code and the UN Model Regulations. According to the document, the Sub-Committee should request the Secretariat to bring this issue to the attention of the UN Sub-Committee of Experts for the Transport of Dangerous Goods and ask for clarification. The Sub-Committee should decide:

.1 whether special provision 132 should be adopted in the IMDG Code;

.2 whether stowage code SW11 should be included in column 16a; or

.3 whether both amendments should be done,

and instruct the E&T Group accordingly.

6.73 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 SW1 is already allocated for UN 2956;

.2 the Secretariat should be requested to inform UNTDG on this matter;

.3 P409 also applies to UN 3242 and UN 3251, and for these, transport in limited quantities is permitted; and SP132 had been previously allocated for UN 2956, along with SP133; however, SP132 was removed (DSC 17/3, paragraph 3.41); and at that time, the Sub-Committee did not consider adding SW11 for unknown reasons; therefore, adding SW11 can be supported; and

.4 the harmonization can be supported, and therefore both proposed amendments should be made.

6.74 After consideration, the Sub-Committee agreed to;

.1 refer the document to E&T 38 for further consideration and incorporation, as appropriate, into draft amendment 42-24 to the IMDG Code; and

.2 instruct the Secretariat to bring this issue to the attention of UNTDG after consideration at E&T 38.

Stowage of n.o.s. substances of class 4.3

6.75 The Sub-Committee had for its consideration document CCC 8/6/9 (Germany), proposing to amend the stowage category from E to D in the Dangerous Goods List for UN 3129, UN 3130 and UN 3148, and to require "stowage protected from sources of ignition", in order to facilitate compliance with SOLAS regulation II-2/19 and the stowage requirements of the IMDG Code for these UN numbers.

6.76 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 It is important for the carrier to be informed of flashpoint. Taking this into account, the proposal may not be appropriate. In this context, there are two possible solutions. UNTDG could be requested to consider the establishment of new general entries for class 4.3 with subsidiary hazard of class 3, and other entries, if necessary. If these entries are assigned to the goods, the shipper shall indicate flashpoint on the transport document in accordance with 5.4.1.4.3.6 of the IMDG Code. Alternatively, a new maritime special provision could be created in order to require class 3 hazard communication, such as documentation, labelling and placarding, if the flashpoint is less than 23°C, and assign it to UN 3129, 3130 and 3148.

.2 UNTDG should be tasked to create a separate entry first.

.3 It is not self-evident that substances of class 4.3, subsidiary hazard class 3, with flashpoint less than 60°C, would be transported under the correct conditions.

.4 Administrative burden should be taken into account.

.5 Stowage requirements should be amended now, even if UNTDG will be informed.

6.77 After consideration, the Sub-Committee agreed to:

.1 refer the document to E&T 38 for further consideration; and

.2 instruct the Secretariat to bring this issue to the attention of UNTDG after consideration at E&T 38, if appropriate.

Harmonization of SP388 with the UN Model Regulations and further consequential amendments to SP961.1 and SP962.4

6.78 The Sub-Committee had for its consideration document CCC 8/6/10 (Germany), proposing an amendment to SP388 to harmonize the provision with the UN Model Regulations, as well as other consequential amendments to SP961.1 and SP962.4, closing a regulatory gap in the IMDG Code, as the testing requirements for lithium batteries in provision 2.9.4

currently apply only to hybrid electric vehicles loaded in the cargo spaces of a ro-ro ship, but not to other hybrid electric vehicles to which SP961.2 through SP961.4 applies.

6.79 Following the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 SP961.1 allows an exemption from the provisions of the IMDG Code if vehicles are stowed in, e.g. ro-ro spaces. The IMDG Code specifies the requirements for lithium batteries only in SP961.1 and SP962.4, but not in SP388. Paragraph 3 of document CCC 8/6/10 states that hybrid electric vehicles containing a lithium battery are exempted if the liquid fuel has a high flashpoint or the fuel tank is empty. However, if those vehicles are transported by other than ro-ro ships, such vehicles cannot be exempted from the provisions of the IMDG Code and should be transported as UN 3166 VEHICLE, FLAMMABLE LIQUID POWERED. Therefore, the proposal is not supported.

.2 The proposal can be supported.

6.80 After consideration, the Sub-Committee agreed to the proposal, in principle, and decided to refer the document to E&T 38 for consideration and incorporation, as appropriate, into draft amendment 42-24 to the IMDG Code.

Stabilized substances

6.81 The Sub-Committee had for its consideration document CCC 8/6/11 (Morocco et al.), setting forth proposals to amend the IMDG Code to address identified deficiencies regarding special transport conditions and operational controls in order to ensure safe transport. In order to ensure greater visibility into cargo transported at a temperature higher than ambient, the co-sponsors propose to amend SP386.

6.82 During the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 Document CCC 5/6/10 (ICS) contained a similar proposal. The Sub-Committee did not agree to the proposal (CCC 5/13, paragraph 6.35). There is no justification to agree to the proposal at hand, either.

.2 The shipper's existing requirements should be taken into account.

.3 This matter should be brought to the attention of UNTDG first. SAPT is used to define whether a monomer is a class 4.1 polymerizing substance, and whether temperature control is required. Only in the latter case an exact SAPT value is required and the words "TEMPERATURE CONTROLLED" shall be added as part of the proper shipping name. Beyond that, there is no practical need for the SAPT in defining safe handling conditions. Monomer polymerization and stability is far too complex a process to be reduced to one simple temperature value and there is even potential for misapplication resulting in dangerous consequences. Prevention of dangerous reactions in stabilized monomers is achieved by addition of chemical inhibitors which ensure that the monomer remains stable for the duration of transport. However, the absolute value of both the SAPT and the duration of stabilization can change with age, headspace, fill level and many other variables. Therefore, communicating the SAPT, whether an exact value or that the SAPT is above the temperature control threshold, would not improve the safe transport of monomers. Moreover, the proposal to require inclusion of "the anticipated duration of the effectiveness of inhibitors" in the transport document is unnecessary, as the IMDG Code already requires the shipper to certify each shipment to be in "accordance with the applicable regulations", including SP386.

.4 The proposal can be supported but needs some further consideration.

.5 Substances stabilized by temperature control are already required to have their control and emergency temperatures indicated in the dangerous goods transport document. Furthermore, the IMDG Code already imposes stowage, handling and segregation provisions to stabilized substances that have SP386 assigned. Including additional communication on special transport conditions or operational controls in the dangerous goods transport document could be redundant and lead to conflict or misinterpretation of the provisions of the Code. In addition, the proposal has multimodal implications, and therefore it should be considered at UNTDG first.

.6 This matter was not completely solved at CCC 5 and should be considered further.

6.83 After consideration, the Sub-Committee agreed to refer this matter to E&T 38 for further consideration, with a view to providing further advice to CCC 9.

Proposed amendments to the shipping conditions of seed cake in the IMDG Code

6.84 The Sub-Committee had for its consideration document CCC 8/6/12 (China), proposing amendments to the shipping conditions of seed cake in the IMDG Code. In chapter 3.3 of the IMDG Code, it is proposed to include a new special provision 9XX and assign it to UN 1386(a), UN 1386(b) and UN 2217.

6.85 During the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 the proposed exemptions should include the need for evidence of the stated oil and/or moisture contents of the cargo being shipped, as is the case in the IMSBC Code;

.2 the proposed amendments should only be considered after the conclusion of discussions on 900 series of special provisions;

.3 this matter should be considered at UNTDG first;

.4 a requirement for a certificate from the competent authority should be considered; and

.5 exemptions should be based on testing.

6.86 After consideration, the Sub-Committee agreed to invite the delegation of China to submit a new proposal on this matter to CCC 9, taking into account the comments made above.

The potential hazard of silicon slurry (kerf loss slurry from silicon ingot slicing) during maritime transport

6.87 The Sub-Committee had for its consideration documents CCC 8/6/13 and CCC 8/6/14 (China), proposing the Sub-Committee to note the potential hazard associated with the

maritime transport of silicon slurry, and consider inviting interested Member States and international organizations to participate in an informal group for further research in this aspect, with a view to identifying the hazard characteristics of silicon slurry and clarifying the conditions for safe transport; and introducing the experiments conducted by China on the hazard characteristics of silicon slurry (kerf loss slurry from silicon ingot slicing).

6.88 After consideration, the Sub-Committee agreed to invite interested Member States and international organizations to contact the delegation of China* for further research, with a view to identifying the hazard characteristics of silicon slurry and clarifying the conditions for safe transport.

Proposed amendment to special provision 964 of the IMDG Code

6.89 The Sub-Committee had for its consideration document CCC 8/6/15 (Chile), containing a proposal to amend SP964 of the IMDG Code with the aim of aligning that provision with the current tests in the UN *Manual of Tests and Criteria* and to provide consistency in the description and requirements between the IMDG and IMSBC Codes.

6.90 During the discussion, the Sub-Committee noted the following views expressed on this matter:

.1 the proposal can be supported;

.2 substances that pass the appropriate test in the UN *Manual of Tests and Criteria* can already be exempted from the provisions of the IMDG Code regardless of their particle size; hence, the proposal is not supported;

.3 this matter should be brought to the attention of UNTDG first;

.4 it should be noted that particle size is only a reference;

.5 the proposed amendments should be considered after the conclusion of discussions on 900 series of special provisions; and

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.6 there is no need for harmonization with the IMSBC Code; and exemptions should be based on testing, and both O.1 test and O.3 should be used in this regard.

6.91 After consideration, the Sub-Committee agreed to refer this matter to the Drafting Group for further consideration, time permitting.

Safety concerns arising from fire incidents on board container ships carrying dangerous goods

6.92 The Sub-Committee noted with appreciation document CCC 8/INF.13 (Singapore), reporting on the safety concerns arising from fire incidents on board container ships carrying dangerous goods.

ESTABLISHMENT OF A DRAFTING GROUP

6.93 Having considered the above matters, the Sub-Committee established the Drafting Group on Amendments to the IMDG Code and instructed it, taking into account the comments made and decisions taken in plenary, to:

w*ith regard to devices in use or intended for use during transport containing dangerous goods:*

.1 based on document CCC 8/6/4, prepare draft amendments to provision 5.5.4 of the IMDG Code, for incorporation, as appropriate, into draft amendment 42-24;

with regard to the relevant parts of documents CCC 8/6/1, CCC 8/6/17 and CCC 8/INF.16:

.2 further develop amendments to the IMDG Code concerning:

.1 CARBON; and

.2 competent authorities in terms of certification, approvals and permits;

with regard to other related issues concerning maritime special provisions:

.3 taking into account the relevant parts of documents CCC 8/6/1 and CCC 8/6/16, consider possible consequences for the IMSBC Code;

with regard to special provisions for the transport of vehicles:

.4 prepare draft terms of reference for a correspondence group, taking into account document CCC 8/6/6; and

with regard to special provision 964 of the IMDG Code:

.5 further consider the proposal contained in document CCC 8/6/15, if time permits.

Report of the Drafting Group on Amendments to the IMDG Code

6.94 Having considered the report of the Drafting Group on Amendments to the IMDG Code (CCC 8/WP.7), the Sub-Committee approved it in general and took action, as described in the following paragraphs.

[to be prepared by the Secretariat in consultation with the Chair after the session, based on the Group's report and the actions requested therein, taking into account the decisions taken by the Sub-Committee during subsequent discussions.]

DRAFT AMENDMENT 42-24 TO THE IMDG CODE AND INSTRUCTIONS TO THE E&T GROUP

Instructions to the E&T Group

6.95 The Sub-Committee authorized E&T 38, taking place in the spring of 2023, to prepare draft amendment 42-24 to the IMDG Code, based on documents submitted to CCC 8 and taking into account comments made and decisions taken in plenary. E&T 38 should take into consideration the outcome of the UNSCETDG with regard to the amendments to the twenty-second revised edition of the UN Recommendations on the Transport of Dangerous Goods, Model Regulations. The Group should also identify and correct any editorial mistakes in amendment 41-22 of the IMDG Code and submit a written report to CCC 9.

7 AMENDMENTS TO THE INTERNATIONAL CODE FOR THE SAFE CARRIAGE OF GRAIN IN BULK (RESOLUTION MSC.23(59)) TO INTRODUCE A NEW CLASS OF LOADING CONDITIONS FOR SPECIAL COMPARTMENTS

Background

7.1 The Sub-Committee recalled that MSC 101 had agreed to include, in the 2020-2021 biennial agenda of the Sub-Committee and the provisional agenda of CCC 7, an output on "Amendments to the International Code for the Safe Carriage of Grain in Bulk (resolution MSC.23(59)) to introduce a new class of loading conditions for special compartments", with a target completion date of 2020.

7.2 The Sub-Committee also recalled that MSC 102 had adjusted the target completion year to 2021 due to the postponement of CCC 7. Due to the COVID-19 pandemic, this agenda item was postponed again to CCC 8.

Amendments to the Grain Code

7.3 The Sub-Committee noted annex 3 to document MSC 101/21/12 (Canada and the United States), providing the draft amendments to the *International Code for the Safe Carriage of Grain in Bulk* in order to introduce a new class of loading conditions for "specially suitable compartment, partly filled in way of the hatch opening, with ends untrimmed" and specify the requirements under which grain may be carried in such compartments.

Establishment of the Drafting Group

7.4 The Sub-Committee agreed to establish the Drafting Group on Amendments to the Grain Code and instruct it, taking into account the comments made and decisions taken in plenary, to:

.1 prepare the amendments to the Grain Code based on annex 3 to document MSC 101/21/12, including any consequential modifications and to rectify any editorial mistakes; and

.2 prepare the check/monitoring sheet in accordance with the *Guidance on drafting of amendments to the 1974 SOLAS Convention and related mandatory instruments* (MSC.1/Circ.1500/Rev.1).

Report of the Drafting Group

7.5 Having considered the report of the Drafting Group on Amendments to the Grain Code (CCC 8/WP.6), the Sub-Committee approved it in general and took action, as described in paragraphs 7.6 to 7.[…].

[to be prepared by the Secretariat in consultation with the Chair after the session, based on the Group's report and the actions requested therein, taking into account the decisions taken by the Sub-Committee during subsequent discussions]

8 REVISION OF THE REVISED RECOMMENDATIONS FOR ENTERING ENCLOSED SPACES ABOARD SHIPS (RESOLUTION A.1050(27))

Background

8.1 The Sub-Committee recalled that MSC 101 had agreed to include, in the 2020-2021 biennial agenda of the Sub-Committee and the provisional agenda of CCC 7, an output on "Revision of the Revised recommendations for entering enclosed spaces aboard ships (resolution A.1050 (27))", with a target completion date of 2020.

8.2 The Sub-Committee also recalled that MSC 102 had adjusted the target completion year to 2021 due to the postponement of CCC 7. Due to the COVID 19 pandemic, this agenda item was postponed again to CCC 8.

Proposed amendments in relation to the term "cargo compressor room"

8.3 The Sub-Committee had for its consideration the following documents:

.1 CCC 8/8 (SIGTTO), proposing to delete the term "cargo compressor room" from the list of examples in paragraph 2.1 of resolution A.1050(27), in particular, that while not an "enclosed space" there are risks associated with entry into the cargo compressor room; these risks are well understood and mitigated through application of the IGC Code;

.2 CCC 8/8/1 (China), commenting on why "cargo compressor room" should not be deleted and proposing to simplify the procedures for entering the cargo compressor room; and

.3 CCC 8/8/2 (INTERMANAGER, et al.), commenting on document CCC 8/8 and proposing recommendations relating to the cargo compressor room and revision to resolution A.1050(27), featuring unintended consequences and providing high level information gathered on matters relating to enclosed space events.

8.4 In the ensuing discussion, the Sub-Committee noted the following views:

.1 the cargo compressor room should continue to be considered as an enclosed space for safety reasons, and therefore it should not be deleted from the definition of enclosed space in paragraph 2.1 of resolution A.1050(27);

.2 the removal of cargo compressor rooms from the list of examples could lead the reader to not consider making an assessment for these spaces, which would result in the absence of safety safeguard measures such as supervision, communication and rescue for personnel entering the cargo compressor room;

.3 taking into account the number of incidents and enclosed space-related fatalities, cargo compressor rooms should not be removed from the list of examples and a cautious approach should be taken;

.4 a space designed or required to only be entered during normal cargo handling operations cannot be considered to be a space designed for continuous worker occupancy;

.5 paragraphs 12.1.1 and 17.21.6 of the IGC Code classified the cargo compressor rooms as an enclosed space and the proposal in document CCC 8/8/2 cannot be supported;

.6 safe unrestricted access for personnel wearing protective clothing and breathing apparatus is not the same as a room having unlimited (unrestricted) openings for entry and exit; mechanical ventilation being required to run continuously in a space indicates that the space has inadequate ventilation;

.7 due consideration should be given to risks beyond normal cargo handling operations, such as in cases of cargo leakage and mechanical ventilation failure, and concerns should be noted over the risks and hazards to seafarers entering "cargo compressor rooms" on non-IGC Code ships; and

.8 an assessment of whether the recommendations in resolution A.1050(27) should be followed needs to be based on the definition of an enclosed space, not the list of examples.

8.5 Following discussion, the Sub-Committee noted that there was not sufficient support for deletion of the term of "cargo compressor rooms" as proposed in document CCC 8/8, while there was also no consensus on the alternative approach proposed in document CCC 8/8/2. In this context and having also noted that MSC 106 had received a proposal for a new output on "Revision of resolution A.1050(27)" (document MSC 106/16/1 (China)), the Sub-Committee agreed that no specific actions needed to be taken by this session under this agenda item and that the agenda item should be kept in abeyance, pending the decision from the Maritime Safety Committee on the proposed new output. In addition, the Sub-Committee noted that documents CCC 8/8, CCC 8/8/1 and CCC 8/8/2, together with document CCC 8/5/11 (see also paragraph 5.[…]), may be further considered at a future session, if the new output is agreed by the Committee.

Lessons learned and preventive measures identified from accidents involving entering enclosed spaces aboard

8.6 The Sub-Committee noted that III 8 had invited CCC 8 to consider document III 8/4/5 (China) on Lessons learned and preventive measures identified from accidents involving entering enclosed spaces aboard in relation to the existing output "Revision of the Revised recommendations for entering enclosed spaces aboard ships (resolution A.1050(27))" (III 8/WP.1/Rev.1, paragraph 4.13.13).

8.7 Having recalled the decision in paragraph 8.5 above, the Sub-Committee agreed to further consider document III 8/4/5 at CCC 9.

9 CONSIDERATION OF REPORTS OF INCIDENTS INVOLVING DANGEROUS GOODS OR MARINE POLLUTANTS IN PACKAGED FORM ON BOARD SHIPS OR IN PORT AREAS

General

9.1 The Sub-Committee recalled that MSC 105 approved MSC.1/Circ.1649 on *Guidelines for the implementation of inspection programmes for cargo transport units* and encouraged Member States and international organizations to implement them.

9.2 The Sub-Committee also recalled that CCC 7, having considered the results of container inspection programmes based on submissions from Member States, had expressed its appreciation to them and requested them to continue to submit such reports in accordance with MSC.1/Circ.1442, as amended by MSC.1/Circ.1521.

9.3 The Sub-Committee further recalled that the Secretariat had developed a GISIS functionality to allow Member States to submit the reports of inspection programmes online through the GISIS platform, as published in Circular Letter No.3844 of 8 May 2018.

Inspection programmes for cargo transport units carrying dangerous goods

9.4 The Sub-Committee noted document CCC 8/INF.2 (Secretariat), containing the consolidated results of container inspection programmes, submitted by Chile, Finland, the Republic of Korea and the United States, through GISIS.

9.5 The Sub-Committee was informed that, out of the 64,283 CTUs inspected, 3797 CTUs were found to have deficiencies, which means that 5.91% of the CTUs inspected had deficiencies. With regard to the type of deficiencies, placarding and marking of CTUs, securing/stowage inside the unit, marking and labelling of packages, documentation and serious structural deficiencies were the main deficiencies found.

9.6 The Sub-Committee expressed its appreciation to those Member States that had submitted the results of container inspection programmes and urged Member States who have not yet carried out container inspection programmes to submit the relevant information to the Organization in accordance with MSC.1/Circ.1649.

10 REVIEW OF THE IGC CODE

Background

10.1 The Sub-Committee recalled that MSC 103 agreed to include in its post-biennial agenda an output on "Review of the IGC Code", with two sessions needed to complete the item, assigning the CCC Sub-Committee as the associated organ. Having agreed that the scope of the review should not be limited to specific paragraphs of the Code, the Committee also agreed, in accordance with MSC.1/Circ.1481 and MSC.1/Circ.1500/Rev.1, that:

.1 the amendments to be developed should apply to all new ships to which the IGC Code applied on or after the date of entry into force;

.2 the instrument to be amended was the IGC Code; and

.3 the amendments to be developed should enter into force on 1 January 2028, provided that they were adopted before 1 July 2026.

New product for inclusion in chapter 19 of the IGC Code

10.2 The Sub-Committee considered document CCC 8/10 (Norway), providing the background and details of a new product, "VOC Condensate", for inclusion in chapter 19 of the IGC Code. Following discussion, the Sub-Committee instructed the Working Group on Amendments to the IGF Code and Review of the IGC Code to further consider document CCC 8/10, with a view to preparing amendments to the IGC Code.

Proposals related to amendments to the IGC Code

10.3 The Sub-Committee had for its consideration the following documents:

.1 CCC 8/10/1 (SIGTTO), proposing to amend the IGC Code to correct the stated triple point for CO_2 in paragraph 17.21.1 and designate CO_2 as toxic in chapter 19 and proposing to amend the classification of "Carbon dioxide (high purity)" and "Carbon dioxide (reclaimed quality)" to include "toxic" in addition to "asphyxiant";

.2 CCC 8/10/12 (Japan), commenting on document CCC 8/10/1 on amendments to the IGC Code to correct the stated triple point for CO_2 in

paragraph 17.21.1 and designate CO_2 as toxic in chapter 19, in particular that a thorough deliberation is necessary for the amendment to chapter 19;

.3 CCC 8/10/2 (SIGTTO), proposing to amend the IGC Code to align it with the latest industry best practice and guidance on ESD systems, in particular to retain the valve location requirements in section 5.5;

.4 CCC 8/10/3 (IACS and SIGTTO), providing draft amendments of the IGC Code, incorporating published relevant unified interpretations;

.5 CCC 8/10/4 (IACS), providing draft amendments to the IGC Code, incorporating draft unified interpretations submitted to the previous session of the Sub-Committee and additional ones developed by IACS since then; further draft amendments of the IGC Code incorporating three new draft unified interpretations developed by IACS since CCC 7; and draft amendments to the IGC Code incorporating draft unified interpretations presented in documents CCC 7/11 and CCC 7/11/5;

.6 CCC 8/10/6 (IACS), seeking clarification from the Sub-Committee regarding the certification requirements of gas carriers when using a low flashpoint fuel (LFPF) system, completely independent of the cargo systems, in particular regarding the transparency of the CoF applicable to a gas carrier, which is to be fitted with an LFPF system, totally independent of its cargo system; and that several amendments to appendix 2 "Model Form of International Certificate of Fitness for the Carriage of Liquefied Gasses in Bulk" of the IGC Code are proposed;

.7 CCC 8/10/7 (Japan, Norway and SIGTTO), proposing an amendment to the IGC Code with regard to the scope of application of the requirement for piping system testing, in particular that the requirement for strength test need not be applied to all piping and paragraph 5.13.2.1 of the IGC Code should be amended;

.8 CCC 8/10/8 (Japan or Norway), proposing an amendment to the IGC Code with regard to requirements related to the reference temperature in chapter 15, in particular that the prevention of liquid-full in cargo tanks is

necessary for the safety of a cargo tank and thus amendment to the IGC Code is necessary;

.9 CCC 8/10/9 (Japan and Norway), proposing an amendment to the IGC Code with regard to the ventilation requirement for the space not normally entered, in particular that the required ventilation rate may be reduced for enclosed spaces not normally entered, as defined in paragraph 12.2 of the previous Code and that paragraph 12.2 of the current Code should be amended and two examples are also provided;

.10 CCC 8/10/10 (China), proposing an amendment to the IGC Code, to modify the new regulation 16.6, to add main steam valve protection requirements of steam propulsion devices to reduce the risks of failure of the main steam valve caused by improper operation; and

.11 CCC 8/10/11 (China), providing suggestions on the revision of the IGC Code by introducing the significance of the meaning "gastight" and proposing a draft guideline on tests for gastight in order to ensure the effective implementation of the gastight requirements.

10.4 Following discussion, the Sub-Committee noted the following views expressed:

.1 in addition to the submissions to this session, the documents submitted to MSC 102, i.e. documents MSC 102/21/1, MSC 102/21/14, CCC 8/10/12, MSC 102/21/16 and MSC 102/21/20, should also be taken into account by the Working Group;

.2 the proposals in document CCC 8/10/1 to amend the IGC Code to correct the stated triple point for CO_2 in paragraph 17.21.1 and designate CO_2 as toxic in chapter 19 should be carefully considered, in particular on the possible impact on fire protection systems and operational aspects for the crew; there should be a clear distinction between the CO_2 carried as a cargo and the CO_2 involved in the insulations for the safety of ships;

.3 with regard to document CCC 8/10/10, new steam propulsion installations are unlikely on new vessels and, therefore, the application scope should be carefully considered;

.4 for document CCC 8/10/11, a definition for gastight in the IGC Code should be agreed first and then a consistent approach is needed in the SOLAS Convention, the IBC Code, the OSV Chemical Code, etc.;

.5 regarding document CCC 8/10/9, if option 2 for amendments to paragraph 12.2 of the IGC Code is preferred, the end of the proposal should read as "safe environment can be ensured before entering the spaces"; and

.6 documents CCC 8/10/1 and CCC 8/10/12 should be carefully considered in the Working Group, without initiating amendments to the IGC Code.

10.5 Taking into account the above views, the Sub-Committee decided that the Working Group on Amendments to the IGF Code and Review of the IGC Code, established under agenda item 3 (see paragraph 3.[…]), should be instructed to:

.1 further consider documents CCC 8/10/1 and CCC 8/10/12 and advise the Sub-Committee on how best to proceed; and

.2 further consider documents CCC 8/10, CCC 8/10/2, CCC 8/10/3, CCC 8/10/4, CCC 8/10/6, CCC 8/10/7, CCC 8/10/8, CCC 8/10/9, CCC 8/10/10, CCC 8/10/11, MSC 102/21/1, MSC 102/21/14, MSC 102/21/16 and MSC 102/21/20, with a view to preparing amendments to the IGC Code.

Drafting of the safety requirements for use of LPG cargo as fuel

10.6 The Sub-Committee considered documents CCC 8/10/5 and CCC 8/INF.11 (Republic of Korea and United Arab Emirates), proposing the draft of the safety requirements for use of LPG cargo as fuel in relation to the IGC Code (as amended by resolution MSC.370(93)), including information on the result of the study on the use of LPG cargo as fuel from a safety perspective in relation to the IGC Code and proposing a work plan for this agenda item.

10.7 In the ensuing discussion, the Sub-Committee noted the following views expressed:

.1 in addition to addressing the safety requirements for LPG carriers using their cargo as fuel, scenarios such as an LCO_2 (liquefied carbon dioxide) Carrier using LPG or LNG as fuel through a dedicated and independent system (completely separated from the cargo system) should also be considered; and

.2 recognizing the increasing demand for the use of the LPG cargo as fuel in LPG carriers, it is important to develop relevant safety requirements under the IGC Code.

10.8 The Sub-Committee also noted that the discussion on the development of interim guidelines to address safety provisions for ships using LPG fuels under agenda item 3 is still ongoing and the outcomes might have an impact on the safety requirements for use of LPG cargo as fuel in relation to the IGC Code. In this context, the Sub-Committee agreed to hold a decision on documents CCC 8/10/5 and CCC 8/INF.11 in abeyance for a future session, with a view for a correspondence group, if established, to progress the development of safety requirements for use of LPG cargo as fuel in relation to the IGC Code intersessionally.

Instructions to the Working Group

10.9 Having considered the above matters, the Sub-Committee instructed the Working Group on Amendments to the IGF Code and Review of the IGC Code, under agenda item 3 (see paragraph 3.[…]), taking into account the comments made and decisions taken in plenary, to:

.1 further consider documents CCC 8/10/1 and CCC 8/10/12, and advise the Sub-Committee on how best to proceed; and

.2 further consider documents CCC 8/10, CCC 8/10/2, CCC 8/10/3, CCC 8/10/4, CCC 8/10/6, CCC 8/10/7, CCC 8/10/8, CCC 8/10/9, CCC 8/10/10, CCC 8/10/11, MSC 102/21/1, MSC 102/21/14, MSC 102/21/16 and MSC 102/21/20, with a view to preparing amendments to the IGC Code, as well as the associated check/monitoring in accordance with MSC.1/Circ.1500/Rev.1; and

.3 consider whether it is necessary for the Correspondence Group to be re-established and, if so, prepare terms of reference for consideration by the Sub-Committee.

Report of the Working Group

10.10 Having considered the part of the report of the Working Group on Amendments to the IGF Code and Review of the IGC Code (CCC 8/WP.4) related to this agenda item, the Sub-Committee approved it in general and took action as described in paragraphs …

[to be prepared by the Secretariat in consultation with the Chair after the session, based on the Group's report and the actions requested therein, taking into account the decisions taken by the Sub-Committee during subsequent discussions]

11 DEVELOPMENT OF MEASURES REGARDING THE DETECTION AND MANDATORY REPORTING OF CONTAINERS LOST AT SEA THAT MAY ENHANCE THE POSITIONING, TRACKING AND RECOVERY OF SUCH CONTAINERS

Background

11.1 The Sub-Committee recalled that MSC 103 agreed to include in its post-biennial agenda an output on "Development of measures regarding the detection and mandatory reporting of containers lost at sea that may enhance the positioning, tracking and recovery of such containers", with two sessions needed to complete the item, assigning the CCC Sub-Committee as the coordinating organ, in association with the NCSR Sub-Committee as and when requested by the CCC Sub-Committee (MSC 103/21, paragraph 18.34).

Estimate of containers lost at sea and standardized procedure for reporting

11.2 The Sub-Committee had for its consideration the following documents:

.1 CCC 8/11 (WSC), providing updated information from 2020 and 2021 on estimate of containers lost at sea, in particular, that the winter of 2020-2021 saw an unusually high number of incidents. Consequently, when reviewing the results of the 14-year period (2008-2021) surveyed, it was estimated that there were on average a total of 1,629 containers lost at sea each year, which is a significant increase (18%) to the average annual loss for the 12-year period ending in 2019. Notwithstanding this increase, containers lost overboard represent less than one thousandth of 1% (0.001%) of the roughly 241 million packed and empty containers currently shipped each year;

.2 CCC 8/11/1 (Australia et al.), proposing amendments to SOLAS chapter V, with a view to making mandatory reporting of the loss of containers through a standardized procedure. It also includes a consequential amendment to paragraph (3) of Article V (reporting procedures) of Protocol I (incident report) of the MARPOL Convention (Article 8) to avoid double reporting. The submission also considers the issue of how a shipboard fall detection system might assist in such reporting;

.3 CCC 7/14/2 (WSC), containing information regarding WSC's survey of its member companies to accurately estimate the number of containers that are lost at sea each year. For the combined 12-year period from 2008 to 2019, the WSC estimates that there were on average a total of 1,382 containers lost each year;

.4 CCC 7/INF.14 (RINA), summarizing a Workshop held by the RINA Maritime Safety Committee to discuss container ship safety following the submission of CCC 7/14/2. It identifies areas where there were opportunities to improve safety and reduce the increasing numbers of container losses. Further discussion and dialogue with stakeholders are suggested in order to promote the input of naval architects into the wider container ship industry;

.5 CCC 8/INF.7 (the Netherlands), providing information with an aim to support the discussion of the reporting of the loss of containers through a standardized procedure and, in particular, considering the potential use of a shipboard fall detection system which might assist in such reporting; and

.6 CCC 8/INF.9 (France), presenting a study report on technical solutions to locate and monitor containers that have fallen into the sea.

11.3 In the ensuing discussion, the Sub-Committee noted the following views:

.1 mandatory reporting of the loss of containers through a standardized procedure by draft amendments to SOLAS chapter V should be supported;

.2 the reported information should be allowed for further updating and installation of a shipboard system to detect containers lost overboard should

be encouraged, such as mechanical, electronic and/or visual aids for detection of the fall of containers from ships in real time;

.3 ships that have lost containers at sea should report the incidents to the nearest coastal State as well as their own flag States, and subsequently for the flag State to submit report to IMO via GISIS;

.4 the proposed amendments to regulations 31 (Danger message) and 32 (Information required in danger messages) should be further considered in the Working Group;

.5 documents CCC 7/14/2 and CCC 8/11, containing information on several safety improvement initiatives, e.g. amendments to the SOLAS Convention requiring verified gross mass (VGM) of a packed container and MARIN Top Tier Study, should be further considered in the Working Group;

.6 the Legal Committee should also be invited by the Maritime Safety Committee to take appropriate actions on the Wreck Removal Convention (WRC) regarding the loss of containers at sea;

.7 careful technical and legal consideration should be carried out by the Working Group, in particular on the SOLAS regulation I/21 and XI-1/6 with regard to marine casualty;

.8 measures to prevent containers lost at sea and the possible impact of tracking devices on ships' communication systems should also be taken into account; and

.9 documents CCC 8/INF.7 and CCC 8/INF.9 should be further considered in the Working Group.

Establishment of the Working Group

11.4 Following discussion, the Sub-Committee agreed to establish a Working Group on Development of Measures Regarding the Detection and Mandatory Reporting of Containers Lost at Sea and instruct it, taking into account the comments made and decisions taken in plenary, to:

.1 further consider documents CCC 8/11, CCC 8/11/1 and CCC 7/14/2, taking into account documents CCC 7/INF.14, CCC 8/INF.7 and CCC 8/INF.9, with a view to identifying possible measures regarding the detection and mandatory reporting of containers lost at sea that may enhance the positioning, tracking and recovery of such containers;

.2 prepare draft amendments to SOLAS chapter V, based on annex 1 to document CCC 8/11/1;

.3 prepare draft amendments to article V of Protocol I of the MARPOL Convention, based on annex 2 to document CCC 8/11/1; and

.4 consider whether it is necessary to establish the correspondence group and, if so, prepare terms of reference for consideration by the Sub-Committee.

Report of the Working Group

11.5 Having considered the report of the Working Group on Development of Measures Regarding the Detection and Mandatory Reporting of Containers Lost at Sea (CCC 8/WP.5), the Sub-Committee approved it in general and took action as described in paragraphs 11.6 to 11.[…].

[to be prepared by the Secretariat in consultation with the Chair after the session, based on the Group's report and the actions requested therein, taking into account the decisions taken by the Sub-Committee during subsequent discussions]

12 UNIFIED INTERPRETATION OF PROVISIONS OF IMO SAFETY, SECURITY AND ENVIRONMENT-RELATED CONVENTIONS

General

12.1 The Sub-Committee recalled that this is a continuous item on the Sub-Committee's biennial agenda and that the Assembly, at its thirtieth session, expanded the output to include all proposed unified interpretations to provisions of IMO safety, security and environment-related conventions, so that any newly developed or updated draft unified interpretation could be submitted for the consideration of the Sub-Committee, with a view to developing an appropriate IMO interpretation.

Clarification of the IGF Code and IGC Code requirements

Amendments to MSC.1/Circ.1558

12.2 The Sub-Committee considered document CCC 8/12/1 (IACS), proposing draft amendments to sections 1 and 2 of the annex to MSC.1/Circ.1558 on *Unified interpretations of the IGF Code*, to facilitate the consistent and global implementation of paragraphs 2.2.15.3 and 2.2.17 of part A of the IGF Code. Following the discussion, the Sub-Committee noted the following views expressed:

.1 different configurations for the pumps installed in the tank connection space should be taken into account;

.2 where pumps have been included in the design of tank connection spaces, these do not comply with the IGF Code, nor does it comply with the agreed unified interpretations;

.3 there is no definition for the term "pump" in the IGF Code and no agreed unified interpretations; and

.4 if equipment is considered a source of ignition, fire protection measures need to be complied with and such spaces should no longer be classified as a tank connection space but as a fuel preparation room.

12.3 Having noted the views expressed, the Sub-Committee invited IACS to take note of the comments made on the matter and submit a revised proposal to a future session of the Sub-Committee.

Draft unified interpretation of the IGC Code

12.4 The Sub-Committee considered document CCC 8/12/2 (IACS and SIGTTO) proposing a unified interpretation of paragraphs 4.20.3.5, 4.20.3.6, 4.20.3.7, 5.13.2.5 and 13.3.5 of the IGC Code in relation to the conduct of verifications and examinations required during the first full loading and unloading of the cargo. Following the discussion, the Sub-Committee requested the Secretariat to prepare the relevant draft MSC circular (see also paragraph 12.[…]).

Draft unified interpretation on requirements for fuel preparation rooms not located on an open deck

12.5 The Sub-Committee considered document CCC 8/12/3 introducing a draft unified interpretation to provide clarification on applying certain tank connection space requirements to the design of fuel preparation rooms not located on an open deck in compliance with IGF Code section 5.8. Following the discussion, the Sub-Committee noted the following views expressed:

.1 the proposed unified interpretation on requirements for fuel preparation rooms not located on an open deck should be further considered in the Working Group; and

.2 the proposed unified interpretation removes over-interpretations within the IGF Code therefore it is supported.

12.6 Subsequently, the Sub-Committee instructed the Working Group on Amendments to the IGF Code and Review of the IGC Code to further consider document CCC 8/12/3, and advise the Sub-Committee how best to proceed.

Proposed unified interpretation of bunkering manifold arrangements fitted on LNG bunkering ships

12.7 The Sub-Committee considered document CCC 8/12/5 (IACS) discussing matters identified with new building ships designed to be LNG bunkering ships, which, due to their operational particularities, are fitted with cargo transfer equipment in addition to the traditional cargo manifolds; the document proposes a draft unified interpretation to address those issues. Following the discussion, the Sub-Committee noted the views expressed that the proposal in document CCC 8/12/5 was a substantial matter leading to additional requirements on ships and, therefore, it should be considered in the review of the IGC Code.

12.8 Subsequently, the Sub-Committee instructed the Working Group on Amendments to the IGF Code and Review of the IGC Code to further consider document CCC 8/12/5 and advise the Sub-Committee how best to proceed.

Lashing software as a supplement to container stowage and securing plan

12.9 The Sub-Committee considered document CCC 8/12 (IACS) discussing the acceptance of lashing software as a supplement to the stowage and securing plan included in

the approved Cargo Securing Manual (CSM), in order to evaluate actual loading conditions, and proposing a draft unified interpretation for endorsement by IMO. Following the discussion, the Sub-Committee noted the following views expressed:

.1 international harmonization and promotion of the use of lashing software should be encouraged;

.2 a careful consideration of the possible effects on Administrations, shipowners and companies, including the need for performance standards on the lashing software should be performed;

.3 in the case of a new approval process of lashing computer software by Administrations or recognized organizations, the application date and relevant administrative burdens should be taken into account;

.4 the proposed interpretation has a significant impact on the industry, as the interpretation requires that the software be approved by the Administration;

.5 the application of loading software to containers by extending section 3 of the MSC.1/Circular.1353/Rev.2 on *Revised Guidelines for the Preparation of the Cargo Securing Manual* to cargoes in section 4, goes beyond an interpretation and, instead, the guidance should be reviewed to consider the appropriateness of using software to supplement or replace steps in the stowage and securing of containers;

.6 the criteria for approval of lashing software should be carefully considered and the Sub-Committee should invite interested Member States to submit proposals for a new output; and

.7 a harmonized performance standard for lashing software should be developed.

12.10 Noting that the proposal in document CCC 8/12 was beyond the remit of a unified interpretation, the Sub-Committee invited interested Member States and international organizations to submit proposals for a new output on the matter on lashing software as a supplement to the container stowage and securing plan.

Draft MSC circulars on unified interpretations of the IGC and IGF Codes

12.11 Having considered the above documents and the draft UIs agreed in plenary during this session (CCC 8/WP.8), the Sub-Committee agreed to the consolidated draft MSC circulars on unified interpretations of the IGC Code, as set out in annex […], and on the unified interpretation of the IGF Code (see also paragraph 3.[…]), as set out in annex […], with a view to approval by MSC 107 (see also paragraph 3.[…]).

13 DEVELOPMENT OF GUIDELINES FOR THE SAFETY OF SHIPS USING AMMONIA AS FUEL

Background

13.1 The Sub-Committee recalled that CCC 7 agreed to hold the development of guidelines for the safety of ships using ammonia as fuel in abeyance, while waiting for the decision of MSC 104 on this matter and to instruct the Correspondence Group on Amendments to the IGF Code and Development of Guidelines for Low-flashpoint Fuels to collect information on the safety of ships using ammonia as fuel, taking into account documents CCC 7/3/9 and CCC 7/INF.8.

13.2 The Sub-Committee also recalled that MSC 105 agreed to include in the biennial agenda of the CCC Sub-Committee for 2022-2023 and the provisional agenda for CCC 8 an output on "Development of guidelines for the safety of ships using ammonia as fuel", with a target completion year of 2023 (MSC 105/20, paragraph 18.26).

Report of Correspondence Group and development of guidelines

13.3 The Sub-Committee considered the report of the Correspondence Group on Development of Technical Provisions for the Safety of Ships using Low-flashpoint Fuels, as set out in document CCC 8/13 (Japan), regarding the collection of safety information for the use of ammonia, including the list of safety information for ammonia as a fuel; the list of items to be considered for the development of safety provisions and the list of scenarios related to the items; and noted a summary of comments provided to the Correspondence Group, regarding collection of the safety information for the use of ammonia as contained in document CCC 8/INF.10 (Japan).

13.4 Having approved the report of the Correspondence Group in general, the Sub-Committee also noted:

.1 the discussion of the Correspondence Group on the collection of information on the safe use of ammonia (CCC 8/13, annex 1, paragraph 5);

.2 the discussion of the Correspondence Group on preparation for the development of guidelines for the safety of ships using ammonia fuel (CCC 8/13, annexes 2 and 3, paragraph 6 to 10); and

.3 the outcomes of the Correspondence Group to discuss for the development of guidelines for the safety of ships using ammonia fuel (CCC 8/13, paragraph 11).

13.5 The Sub-Committee also had for its consideration the following documents:

.1 CCC 8/13/1 (Japan, Singapore, ICS and INTERCARGO), providing information on possible issues to be considered for developing guidelines for the safety of ships using ammonia as fuel and proposing the way forward; and

.2 CCC 8/13/2 (Republic of Korea), proposing a review of the environmental effect which will be considered in future discussions, in particular, it is believed that effective and meaningful discussions can be conducted only when the safety standards of ammonia and marine pollution standards are reviewed at the same time, and appropriate measures are needed for the development so that the environmental impact of ammonia can be reviewed.

13.6 In the ensuing discussion, the Sub-Committee noted the following views expressed:

.1 document CCC 7/3/9 should be taken into account in the process of the developing guidelines for the safety of ships using ammonia as fuel;

.2 ammonia leakage is the critical risk and consequences to human health and environment should be carefully considered;

.3 development of the safety guidelines should be the priority and it is premature to consider the acceptable emission to the air and the sea at this stage;

.4 due to the toxic nature of ammonia, the "ESD-protected Machinery Spaces" concept should not be used for ammonia and only the "Gas Safe Machinery Spaces" concept should be used;

.5 life-saving appliances and escape routes including those on-deck should be kept a certain distance away from the openings of secondary enclosures and the vent of fuel containment systems;

.6 the available regulations have been validated for use on land for more than 100 years and could be considered to be utilised in future discussions for the development of guidelines for the safety of ships using ammonia as fuel;

.7 the risk of explosion should be assessed before reducing the requirements regarding hazardous area zones and increasing the division of toxic areas;

.8 the environmental impact of ships using ammonia as fuel should be considered in parallel with the safety consideration; and

.9 the operational safety aspect should be properly considered.

Instructions to the Working Group

13.7 Following discussion, the Sub-Committee agreed to instruct the Working Group on Development of Technical Provisions for Safety of Ships using Alternative Fuels, established under agenda item 3 (see also paragraph 3.[…]), taking into account the comments made and decisions taken in plenary, to:

.1 initiate the development of guidelines for the safety of ships using ammonia as fuel, taking into account documents CCC 8/13, CCC 8/13/1 and CCC 7/3/9;

.2 further consider document CCC 8/13/2 with regards to the environmental impact of ships using ammonia as fuel and advise the Sub-Committee how best to proceed; and

.3 consider whether it is necessary to establish the Correspondence Group and, if so, prepare terms of reference for consideration by the Sub-Committee.

Report of the Working Group

13.8 Having considered the relevant part of the report of the Working Group on Development of technical provisions for safety of ships using alternative fuels (CCC 8/WP.3), the Sub-Committee approved it in general and took action as described in paragraphs 13.9 to 13….

[to be prepared by the Secretariat in consultation with the Chair after the session, based on the group's report and the actions requested therein, taking into account the decisions taken by the Sub-Committee during subsequent discussions]

14 REVISION OF THE INTERIM RECOMMENDATIONS FOR CARRIAGE OF LIQUEFIED HYDROGEN IN BULK

Background

14.1 The Sub-Committee recalled that MSC 105 agreed to include in the biennial agenda of the CCC Sub-Committee for 2022-2023 and the provisional agenda for CCC 8 an output on "Revision of the Interim recommendations for carriage of liquefied hydrogen in bulk", with a target completion year of 2024 (MSC 105/20, paragraph 18.28).

Revision of the interim recommendations

14.2 The Sub-Committee had for its consideration the following documents:

.1 CCC 8/14 (Australia and Japan), proposing a revision of the structure of the *Interim Recommendations for carriage of liquefied hydrogen in bulk* (MSC.420(97)), as three parts, i.e. Part A: General (applicable to ships with any type of cargo containment system); Part B: Ships with cargo containment systems of independent cargo tanks using vacuum insulation; and Part C: Ships with cargo containment systems of independent cargo tanks using insulation materials and hydrogen gas in the inner insulation spaces; and

.2 CCC 8/14/1 (Australia and Japan), specifying safety measures for a new liquefied hydrogen carrier through gap analysis of a new liquefied hydrogen carrier and the existing regulations including the IGC Code and the *Interim Recommendations for carriage of liquefied hydrogen in bulk* (MSC.420(97)); hazard identification of a cargo containment systems of a new liquefied hydrogen carrier by the shipyard and the classification society; and

consideration of the possible safety measures by the Administration, the shipyard, the classification society and National Maritime Research Institute.

14.3 In the ensuing discussion, the Sub-Committee noted the following views:

.1 the revision on the *Interim Recommendations for carriage of liquefied hydrogen in bulk* (MSC.420(97)) was needed and documents CCC 8/14 and CCC 8/14/1 should be further considered in the working group; and

.2 a technology-neutral and goal-based approach should be developed in the process of revision, in order to allow the use of all appropriate technologies.

Instruction to the Working Group

14.4 Following discussion, the Sub-Committee agreed to instruct the Working Group on Development of technical provisions for safety of ships using alternative fuels, established under agenda item 3 (see paragraph 3.[…]), taking into account the comments made and decisions taken in plenary, to:

.1 further consider documents CCC 8/14 and CCC 8/14/1, with a view to preparing the draft revised *Interim recommendations for carriage of liquefied hydrogen in bulk* (resolution MSC.420(97)); and

.2 consider whether it is necessary to establish a correspondence group and, if so, prepare terms of reference for consideration by the Sub-Committee.

Report of the Working Group

14.5 Having considered the report of the Working Group on Development of Technical Provisions for Safety of Ships using Alternative Fuels (CCC 8/WP.3), the Sub-Committee approved it in general and took action, as described in paragraphs 14.6 to 14.[…].

[to be prepared by the Secretariat in consultation with the Chair after the session, based on the Group's report and the actions requested therein, taking into account the decisions taken by the Sub-Committee during subsequent discussions]

15 BIENNIAL STATUS REPORT AND PROVISIONAL AGENDA FOR CCC 9

General

15.1 The Sub-Committee noted that MEPC 78 and MSC 105 approved the Sub-Committee's biennial agenda and the provisional agenda for CCC 8 and also agreed to an extension of the Sub-Committee's meeting time to eight days (while maintaining four days interpretation in plenary) for two sessions starting with CCC 8, which has been endorsed by the Council.

15.2 With regard to new outputs, the Sub-Committee noted that MSC 105 had agreed to:

.1 include, in the biennial agenda of the CCC Sub-Committee for 2022-2023 and the provisional agenda for CCC 8 an output on "Development of guidelines for the safety of ships using ammonia as fuel", with a target completion year of 2023;

.2 include, in the biennial agenda of the CCC Sub-Committee for 2022-2023 and the provisional agenda for CCC 8 an output on "Revision of the Interim recommendations for carriage of liquefied hydrogen in bulk", with a target completion year of 2024; and

.3 delete the output on "Amendments to the IMDG Code related to portable tanks with shells made of fibre-reinforced plastics (FRPs) for multimodal transportation of dangerous goods" from the post-biennial agenda of the Committee, as the work under this output had already been covered by amendment 41-22 of the IMDG Code (MSC 105/3/2).

Biennial status report

15.3 Taking into account the progress made at the session, the Sub-Committee prepared the updated biennial status report for the 2022-2023 biennium (CCC 8/WP.2, annex 1), for submission to MSC 106 and MEPC 79 with a view to approval.

Proposed provisional agenda for CCC 9

15.4 Taking into account the progress made at the session, the Sub-Committee prepared the proposed provisional agenda for CCC 9 (CCC 8/WP.2, annex 2), for approval by MSC 106 and MEPC 79.

Correspondence groups established at the session

15.5 The Sub-Committee established correspondence groups on the following subjects, due to report to CCC 9:

[*to be completed by the Secretariat after the session*]

Arrangements for the next session

15.6 The Sub-Committee agreed to establish at its next session working and drafting groups on the following subjects:

[*to be completed by the Secretariat after the session*]

whereby the Chair, taking into account the submissions received on the respective subjects, would advise the Sub-Committee before CCC 9 on the final selection of such groups.

Intersessional meetings

15.7 Having noted that MSC 105 approved the thirty-seventh meeting of the E&T Group (IMSBC Code) to take place immediately after CCC 8, the Sub-Committee invited MSC 106 to approve the holding of two intersessional meetings of the E&T Group for the IMDG Code, one in the spring of 2023 and another one immediately after CCC 9.

Date of the next session

15.8 The Sub-Committee noted that the ninth session of the Sub-Committee has been tentatively scheduled to take place from 20 to 29 September 2023.

[16 ELECTION OF CHAIR AND VICE-CHAIR FOR 2023

16.1 In accordance with the Rules of Procedure of the Maritime Safety Committee and the Marine Environment Protection Committee, the Sub-Committee unanimously re-elected [Ms. MaryAnne Adams (Marshall Islands)] as Chair and [Mr. David Anderson (Australia)] as Vice-Chair, both for 2023.]

17 ANY OTHER BUSINESS

Holistic approach on the human element

17.1 Having recalled that MSC 105 had invited all relevant IMO bodies to assess their respective involvement in the human element within their remit and report back to the

Committee with a view to devising an outline for a holistic approach on the human element, taking into account resource and budgetary implications within the Organization (MSC 105/20, paragraph 16.3), the Sub-Committee invited interested Member States and international organizations to submit relevant proposals, as appropriate, with a view to further consider this instruction at CCC 8.

BoxTech Technical Characteristics Database

17.2 The Sub-Committee noted with appreciation the information contained in documents CCC 8/17 and CCC 7/14 (BIC), providing an update on BIC's progress in deploying the BoxTech Global Container Database, which was launched by BIC to provide a single industry platform for container technical information, including container tare weights needed for method 2 declarations of verified gross mass (VGM) required under SOLAS since 1 July 2016.

ACEP information

17.3 The Sub-Committee considered documents CCC 8/17/1 and CCC 7/14/1 (BIC), providing updated reports on the activity of the Global ACEP Database since CCC 6, in particular that a total of 88 ACEP programmes have now been entered in the database. In this context, the Sub-Committee expressed its appreciation to BIC for its continued commitment to maintaining and running the Global ACEP Database and urged CSC 1974 Contracting Parties to make their ACEP information publicly available and communicate to the Secretariat the location where the ACEP information has been posted, in order for the *List of locations of publicly available ACEP information* (CSC.1/Circ.153), which was approved by MSC 96, to be populated and updated.

Hazard Assessment of Ores and Concentrates for Marine Transport Guidance 2021

17.4 The Sub-Committee note with appreciation the information contained in document CCC 7/INF.11(IIMA), providing information about the publication of updated Guidance on Hazard Assessment of Ores and Concentrates for Marine Transport by the International Council on Mining and Metals (ICMM).

Global Integrated Shipping Information System (GISIS) Database on contact points

17.5 The Sub-Committee considered document CCC 8/17/3 (Secretariat), providing information on the launch of a new GISIS Module on contact points for the IMDG and IMSBC Codes and the CSC Convention. In this context, the Sub-Committee noted that the Secretariat

had developed a new database in the Contacts Point Module of GISIS to collect information and updates regarding the management of Contact points information related to the IMDG and IMSBC Codes and the CSC Convention, aimed at providing Member States with more comprehensive and up-to-date information.

17.6 Furthermore, the Sub-Committee noted that the Secretariat would discontinue amending and issuing MSC.1/Circ.1563/Corr.3, BC.1/Circ.74/Corr.2 and CSC.1/Circ.156/Corr.1, and consequently move the information to the GISIS system. The new GISIS platform would allow Member States to upload and update information related to the IMDG and IMSBC Codes and the CSC Convention for sharing purposes.

UN/OECD seminar

17.7 The Sub-Committee noted with appreciation the information contained in document CCC 8/INF.3 (Secretariat), providing information on the summary and conclusions of the UN/OECD seminar in follow up to the 2020 Beirut port explosion, which was held online on 14 December 2021.

Industry guidelines for warehouses storing, handling and consolidating dangerous goods

17.8 The Sub-Committee noted with appreciation the information contained in document CCC 8/INF.8 (BIC, BIMCO, ICHCA, ICS, IG of P&I Associations, IUMI, IVODGA and WSC), containing industry-produced guidelines in the form of a white paper and accompanying checklist addressing considerations and measures that should be taken by warehouses storing, handling and consolidating dangerous goods, and covers topics such as construction of the warehouse, operations, fire protection, security and emergency response; recommending that guidelines be considered for inclusion in various instruments, codes and circulars published by IMO, as appropriate.

Cost structure of maritime hydrogen transport

17.9 The Sub-Committee noted with appreciation the information contained in document CCC 8/INF.17 (Republic of Korea), providing information on the cost structure of maritime hydrogen transport for the reference of the maritime industry that has been considering alternative fuel that meet carbon emission regulations; analysing the loading port price reflecting the cost of sea transportation of liquefied ammonia and liquefied hydrogen based on the price at the LNG unloading port, and the analysis showed that liquefied ammonia and

liquefied hydrogen were disadvantageous in terms of maritime transportation cost due to calorific value and density per unit.

Group of Experts on the Code of Practice for Packing of Cargo Transport Units

17.10 The Sub-Committee noted with appreciation the information contained in document CCC 8/INF.19 (UNECE Secretariat), providing information on the work being carried out by informal group of experts, further to the request of the UNECE Working Party on Intermodal Transport and Logistics, which had been convened to consider possible amendments to the 2014 edition of the CTU Code.

17.11 In this context, the Sub-Committee noted a statement made by the UNECE Secretariat on the work carried out in the informal pre-work to the Group of Experts on CTU Code, that the informal pre-work was mandated by the UNECE Working Party on Intermodal Transport and Logistics (WP.24) at its sixty-third session in 2020 and further extended at the sixty-fourth session in 2021 in the absence of formal establishment of the Group of Experts on the CTU Code due to lack of necessary decision by the Governing Body of the International Labour Organization (ILO); the informal pre-work so far has resulted in prioritizing specific sections of the CTU Code for updates and a consideration for a number of specific modifications to the CTU Code as indicated in document CCC 8/INF 19 has been worked out; the proposed modifications will be considered by UNECE WP.24 at its sixty-fifth session in October 2022; it is expected that the work on the CTU Code will continue into 2023 either in informal pre-work or in formal group of experts pending a decision by the ILO Governing Body in early 2023; UNECE will report to CCC and/or MSC, as relevant and appropriate, on further progress made on CTU Code updates and decisions made by UNECE WP.24.

Expression of condolence

17.12 The Sub-Committee noted with great sadness the recent passing away of Mr. Christian Breinholt of Denmark, the former Chair of the Maritime Safety Committee from 2012 to 2015. The Sub-Committee expressed its condolences to the Breinholt family and the delegation of Denmark and appreciated his contribution to the work of the Organization and requested the delegation of Denmark to convey its sincere sympathy to his family and colleagues.

17.13 The Sub-Committee also noted with great sadness the recent passing away of Mr. Jack Currie, the long-time delegate to the CCC Sub-Committee, the E&T Group and

UNTDG, representing the International Vessel Operators Dangerous Goods Association (IVODGA). The Sub-Committee appreciated his contribution to the work of the Organization and requested the observer from IVODGA to convey its sincere sympathy to his family and colleagues. A statement from IVODGA is set out in annex ….

Expression of appreciation

17.14 The Sub-Committee expressed appreciation to the following delegates and members of the Secretariat, who had recently relinquished their duties, retired or been transferred to other duties, or were about to do so, for their invaluable contribution to its work and wished them a long and happy retirement or, as the case might be, every success in their new duties:

- Captain Richard Brough (ICHCA International) (on retirement)
- Mr. Hans van der Graaf (IMO) (on retirement)
- Ms. Christine Gregory (IMO) (on retirement)
- Mr. Irfan Rahim (IMO) (on retirement)
- (*please inform the Secretariat if you would like to add a name*)

18 ACTION REQUESTED OF THE COMMITTEES

18.1 The Maritime Safety Committee, at its 106th session, is invited to:

[to be prepared by the Secretariat in consultation with the Chair after the meeting]

18.2 The Marine Environment Protection Committee, at its seventy-ninth session, is invited to:

[to be prepared by the Secretariat in consultation with the Chair after the meeting]

18.3 The Maritime Safety Committee, at its 107th session, is invited to:

[to be prepared by the Secretariat in consultation with the Chair after the meeting]

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ANNEXES

[to be prepared by the Secretariat after the session]